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THE

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VOLUME THE FOURTH.

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ERRATA.

Page 2, col. 1, line 14 from bottom, *for* Palazzo Pitti, *read* Palazzo Vecchio.
— 2, — 2, line 19 from bottom, *for* It is, &c. *read* The Palazzo Vecchio; and line 13 from bottom, *for* the palace, *read* Palazzo Pitti.
— 72, — 2, line 11, *for* Mr. Roscoe, *read* Mr. Horace Smith.
— 167, — 2, Article STEAM NAVIGATION. We have received a pamphlet, entitled *A Brief Narrative, proving the Right of the late* WILLIAM SYMINGTON, *Civil Engineer, to be considered the Inventor of Steam Land Carriage Locomotion, and also of Steam Navigation.* From the statements of the author, Mr. ROBERT BOWIE, and the documents he has produced, it is evident that HENRY BELL availed himself of his knowledge of Mr. Symington's plans, and of an unauthorized inspection of the models of his apparatus, in constructing the boats upon which his unfair claims to the invention were founded.

Mr. JOHN KAY, of Rochdale, has also written, under the impression that Mr. Bell was present when he, Mr. Kay, suggested the idea of working small boats by hand-machinery, and larger vessels by steam-engines. Mr. Kay is, however, evidently mistaken, as he states the conversation to which he alludes to have taken place at Liverpool, in 1796 or 1797, and Mr. Symington had actually worked a boat by steam-machinery in 1788.

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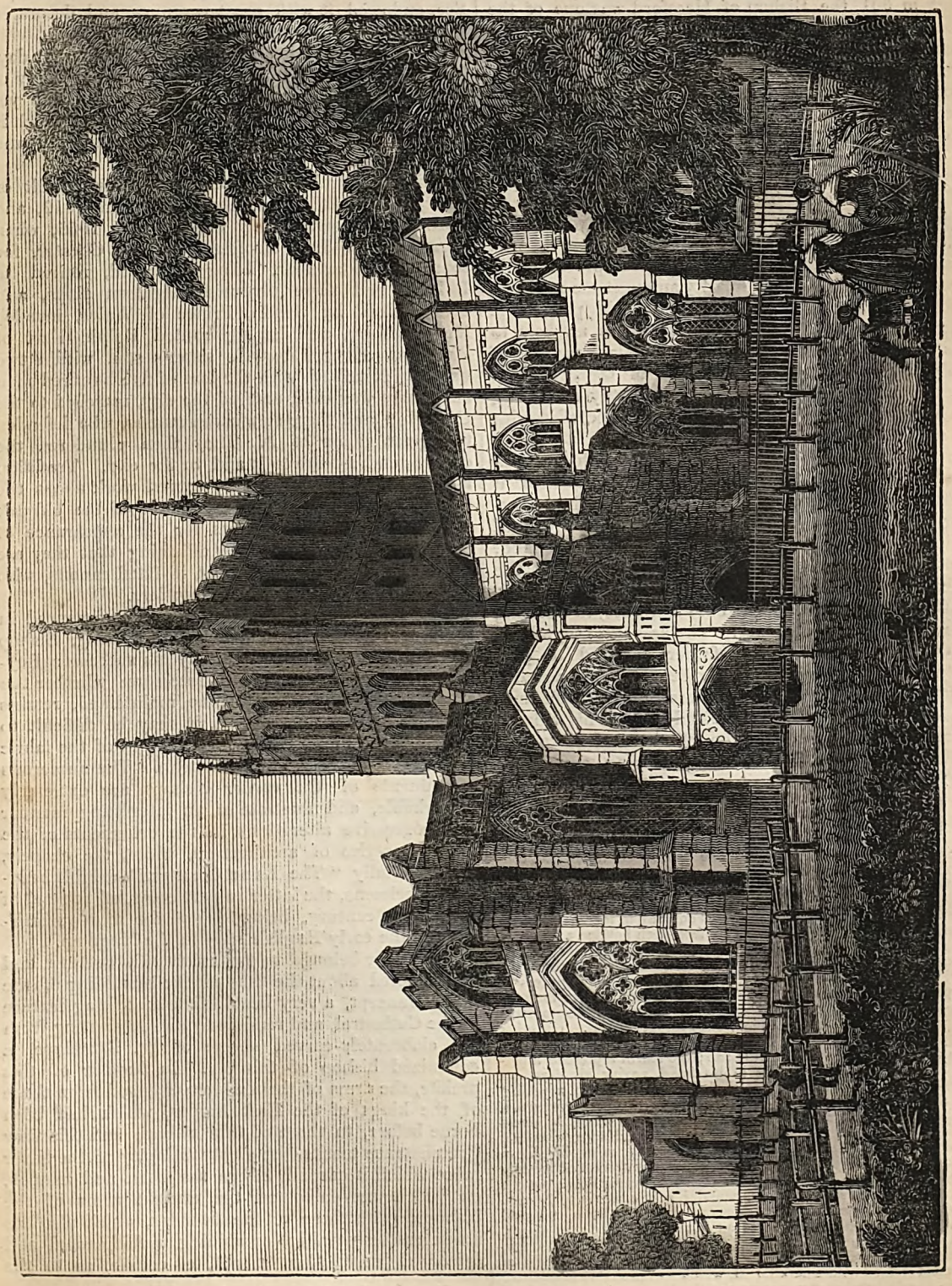

HEREFORD CATHEDRAL.

Vol. V.

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HEREFORD CATHEDRAL.

THE Cathedral Church of St. ETHELBERT, at Hereford, is peculiarly interesting, both on account of its antiquity, and of the numerous architectural beauties which it displays;—not, perhaps, to the cursory inspector: for its general features, though venerable, are rather of a sombre and gloomy character, but it contains curious specimens of almost every style of ecclesiastical architecture, and many valuable and rare appendages.

The See of Hereford is mentioned as Suffragan to the Archbishopric of Caerleon, as early as 544; and in 601, the Bishop of Hereford was one of the seven Saxon prelates who attended the synod convened by Augustine at Canterbury: from which time we trace the regular descent of bishops, in succession, down to the present day. But the See of Hereford becomes more conspicuous in our ecclesiastical history, from a circumstance which is curiously alluded to in a design which forms the embellishments of a remarkably interesting shrine, preserved in the library of the Cathedral*. The circumstance alluded to, is this. Ethelbert, son of Ethelred and Leofrun his wife, succeeded his father in the kingdom of the East Angles, and was a prince of great promise. It seems, that his great popularity amongst his subjects, induced them to desire that he should perpetuate the blessings of his rule, by giving them an heir to his throne: and, accordingly, Althrhida, daughter of Offa, King of the Mercians, and of Quenreda his Queen, was proposed as a princess worthy of sharing in his throne. To gain her hand, he proceeded with a gallant retinue to the Mercian court, at South Town, (now called Sutton Walls, about four miles North of Hereford,) where he was entertained at first with great respect. But Quenreda, either envying his equipage and pomp, or disliking the proposed alliance for her daughter, persuaded her husband, that Ethelbert's views were rather to seduce the affections of his people, than to woo his daughter, and she succeeded, by her base intrigues, in effecting the murder of the Prince. The assassin Guymbert, a domestic of Ethelbert's father, induced to perpetrate this foul deed, by promise of a large reward, conveyed the head of his murdered lord to Offa, and, Judas-like, received the promised price of his villany, upon which Offa directed the remains to be interred, which was done in the neighbouring church at Marden. Being subsequently roused to a sense of his wickedness, by the repeated prodigies which were said to have occurred at the grave, he repented of his crime, and ordered Brithfridus, a Mercian nobleman, to remove the body to the monastery at Hereford, which, with the aid of Egmundus, was accordingly done. Here, say the legends of the day, the miracles were multiplied†, and Milfrid, the Viceroy of Mercia, hearing of their fame, sent thither large sums of money, and caused to be built, some say by the desire of Offa, in expiation of his crime, and in memory of the murdered prince, over his body, a stately and elegant church, appointed it the bishop's See, endowed it with great revenues, and decorated it with splendid ornaments.

Amongst the curiosities shown at the present day, is a fractured, but tolerably well-preserved effigy of Ethelbert, which once stood on a pedestal yet remaining, over against the high altar, and which is said to be the original image of the canonized and patron saint.

This Cathedral of Offa, or Milfrid, was rebuilt or enlarged by Athelstan, about the year 1012, and continued in safety only till the year 1055, when Algar, son of Earl Leofrick, Earl of Chester, for some misdemeanour, being banished, retired to Ireland, and there procuring eighteen ships, and being joined by Griffin, Prince of Wales, invaded Herefordshire, and having routed Ralph, Earl of Hereford, King Edward's sister's son, within two miles of Hereford, with the loss of five hundred men and many wounded, entered the city, marched to the Cathedral, and there slew seven canons, who defended the great doors of it. They then plundered the church, and set fire to it and to the city. Tradition asserts, that the beautifully carved work about the Gothic stalls, now extant, was made of the old oak of that fabric, and accounts for the appearance of *human teeth*, in the wood, by affirming that they are those of the seven noble, but ill-fated canons, who fell a sacrifice to the ruthless invaders.

The rebuilding of the church was commenced in the year 1075, by the pious Robert Lozing, Bishop of Hereford, was carried on by his successor Reynelm, and finished by their successors: whether the south transept, as is by some maintained, retains portions of the early church, it may be difficult to say; but the greater part is composed of the fabric of Athelstan, Lozing, and Reynelm. The numerous monuments of the earlier bishops are extremely remarkable, and illustrate in an interesting manner the progressive alterations in the ornaments of the different styles of architecture, whilst the chapels, and especially "The LADY CHAPEL," exhibit beautiful specimens, the latter in particular, of the end of the twelfth century, being one of the most remarkable of the early English style in the kingdom.

The tower belongs to the thirteenth century, and was erected about the time of the rebuilding of the north transept, a very beautiful and striking feature in the Cathedral, and rendered more interesting from the elaborately-carved shrine of St. Cantelupe, a canonized bishop of this see. In honour of this prelate, the arms of the see were changed from those of the kings of the East Angles, which had been borne before, to those of this bishop; and this very circumstance marks the great antiquity of the silver mace which is carried before the dean and canons, on which are embossed the ancient arms of the bishopric with those of the deanery, which also has an extensive peculiar, combining archidiaconal and episcopal jurisdiction, except, of course, as to the

* This piece of antiquity is formed of oak, so entirely covered with plates of copper, that the wood is no where visible, except at the bottom: it is 8¾ inches high, 7 long, 3½ broad, and much ornamented with enamel and gilding. The colours of the enamel consist of three shades of blue, a green, yellow, white and red. On the front there are two compartments, separated and surrounded by a border. The uppermost contains six human figures, and a kind of bier with a corpse upon it, their heads only are in relief. The two outermost are in long robes, with bare feet, each holding a censer in his hand, the two next are in the act of raising the bier, behind which are two others; one with a pastoral staff seems to direct the way, the other holds a tablet, with the inscription that appears in the plate, representing the shrine, which is certainly older than the time of Henry the Second, and which has been submitted to many learned antiquaries for elucidation, but without success. The other compartment contains four figures, three of them in armour, two with swords, and one with a battle-axe; the fourth represents a person in his robes, with a crown in his hand, paying his devotions before an altar, on which stands a chalice covered with a patine, and a cross on a pedestal. The figure nearest to this personage is cutting off his head, whilst he, in the convulsions of death, seems springing up to meet a hand extended from a cloud, to receive him. At one end, in a Gothic niche, is a figure in long robes, holding a book, and at the other, another *glorified* person in the act of ascending from the earth. The back of the shrine is covered with a pattern of mosaic work in small squares, containing four leaves of flowers. On the ridge of the shrine rises a narrow plate with holes, adorned with three enamelled studs, and on it are three fractured places, from whence some ornaments have been broken off. A red cross, the usual token of a relic, is painted on the inside, on a part of the wood stained with a dark liquid, which was probably considered as the blood of the martyr, and part of the floor on which that blood was shed.

† The water of St. Ethelbert's well, not far from the Cathedral, is even now said to work cures of several maladies.

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especial functions of a bishop, ordination, and confirmation, over thirty-two parishes in and immediately about the city of Hereford.

On the entrance to the Cathedral, its beautiful north porch will attract the attention of the visiter, and in traversing the venerable pile he will be struck with the mutilated appearance of many of the ancient monuments, and by traces of elaborate brasses on the floors. But he may feel a shudder creeping over him, when reminded that these venerable effigies, which were consecrated by the hands of affection and piety, have become the monuments of a blind and misguided zeal, of a sour and uncharitable fanaticism, which not only vented its fury on the images of saints and prelates, but constrains us to mourn the ruin of the once-elegant chapter-house, of part of the cloisters, and many an elaborate and beautiful edifice belonging to this church; not to mention the destruction of two, if not three, of the parish-churches of the city.

But to withdraw the mind from these painful reflections, though not without the hope that the allusion to them will operate as a salutary caution to posterity, let us point out the interesting collection of illuminated manuscripts which the Cathedral library contains. Among them is a copy of Wicliff's Bible; and the valuable and very ancient grants and charters to this church preserved in the Dean's Archive-room. There are also two other curiosities worthy of attention; the one, a very ancient map of the world, drawn with a pen upon vellum stretched on boards; it is five feet four inches wide, by six feet four inches high, and professes to be a copy of the map of Nicodorus, Theodotus, and Polycletus, or Zenodoxus, and to have been drawn by them, as early as the consulate of Julius Cæsar; the triple-mitred prince, the introduction of York Cathedral, and the Norman French, and other such matters, prove it, however, to have been of later date than the Conquest, though it might have been done from the Roman map above alluded to. It was copied a few years ago by the Geographical Society of London, and may, probably, ere long, be given to the public in prints on a reduced scale. It is certainly the greatest curiosity in the kingdom.

The other curious relic was discovered in digging a grave in the Presbyterium of the Choir, in the year 1813, two feet two inches below the marble floor, in a coffin of rude unnailed boards, deposited in the vault with the remains of a corpse, mouldered to dust, except the back part of the skull, on the left side of which was some red hair quite perfect, and a little curled. On the place of the right breast lay the head of the crozier, the staff crossing the body to the left foot. To the crozier was attached, by a skein of silk in a perfect state, the bulla of Pope Clement the Sixth, as the inscription shows: to this, although no traces of it remained, was probably annexed the instrument which had appointed the deceased to the see of Hereford. A hand's breadth below the top of the crozier lay a gold ring, and near it a stone of the amethyst kind, which, on being replaced in the ring, was found to fit exactly. Some pieces of silken texture were spread on the dust, but too much decayed to be removed. It is evident from Leland and other writers, that these relics belonged to John Tullich, forty-eighth bishop of Hereford, who sat there sixteen years, and died about Christmas 1352.

In the year 1786, the massive Norman tower, which stood at the west end of the nave, fell with a tremendous crash, a short time after the congregation had left the Cathedral, and crushed, in its fall, the whole of the upper part of the nave. This part was subsequently rebuilt at a great expense, but unhappily, though under the superintendence of Mr. Wyatt, it was not restored in the original style of the building, and the plain pointed arches above the venerable semicircular ones, based on the noble piers which support them, have not a consistent effect, and are very inferior when compared with what the nave once exhibited, and which still retains its ancient character.

It remains only in this cursory notice of our venerable fabric to observe, that adjoining the Cathedral is the College of Vicars Choral, who form the entire choral force of this Cathedral, being twelve in number, where, excepting the ten choristers and four sub-vicars choral, there are no lay singers. This constitution, though peculiar, has this advantage, that the service of the church is performed by its members in holy orders, and they are well qualified for their interesting duties.

How delightful to hope that those who serve the Lord in this branch of the "beauty of holiness," should, by their sacred calling and their ordination vows, feel not only the responsibility which attaches to them, as the ordained servants of the Lord of life, but, that from this very circumstance, their minds are led to appreciate that far-increased responsibility at the present day, to execute those duties which, when carelessly and coldly performed, justly induce disgust; but when solemnly and feelingly, raise indeed the soul to converse with heaven, attune it to the choirs of saints and angels, enforce the devotion of all who witness it, that "so falling down, they will worship God and confess that God is in us of a truth."

Oh let it not be forgotten, that within these venerable walls, for ages, have the hymns which Miriam sang, and David tuned, the respondent chant and service, and the inspired anthem, floated up to heaven upon the pealing organ's accompaniment, in adoration to Him, to whom for centuries they have been dedicated, and who has deigned to set his name therein. Let it not be forgotten that in such places as this has the daily service been continued, the morning and the evening incense of our church been offered up; let it not be forgotten that when Jehovah ceased to protect *his once favoured people*, when the portentous voice was heard in the temple of Jerusalem, "let us go hence," *their daily service* had ceased, that this was the prelude to their nation's woes. Δ.

SUNDAY is not a day to feast our bodies, but to feed our souls.

THERE is something remarkable in the composition of the Jewish and Christian Scriptures, that, although in every language they are the easiest book to a learner, they are yet dignified, interesting, and impressive. The Pentateuch, Psalms, and Gospels, unite, in a singular degree, simplicity and perspicuity, with force, energy, and pathos. I cannot satisfy myself what the literary peculiarities, the felicities of language are, which make them so universally comprehensible, and yet avoid insipidity, feebleness, and tedium which display so often such genuine eloquence and majesty, and yet are neither affected nor elaborate, nor, in general above the understanding of the common reader.—TURNER'S *History of England.*

IN the time of Henry the Eighth, to be in possession of Tindal's Bible, constituted heresy.—*Life of Henry VIII*

THERE is many a wounded heart without a contrite spirit. The ice may be broken into a thousand pieces, it is ice still. But expose it to the beams of the Sun of Righteousness, and then it will melt.—MIDDLETON.

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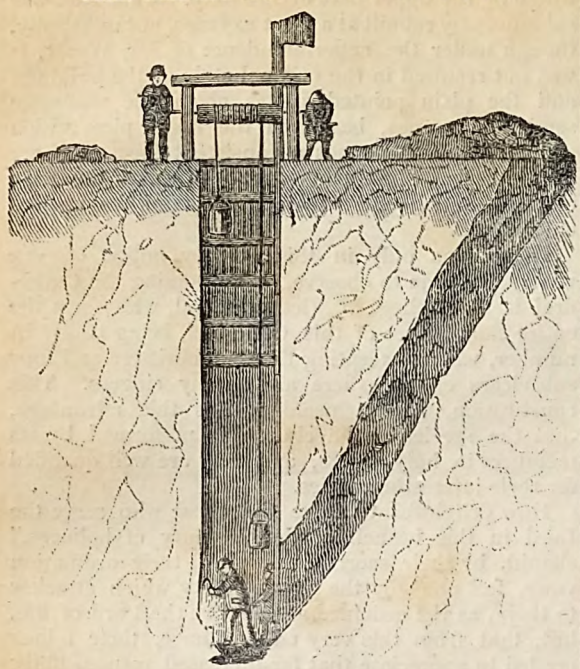
THE MINES OF GREAT BRITAIN.



THE FIRST SHAFT.

No. III. ON THE DISCOVERY OF MINERAL VEINS, AND THE MODE OF FIRST OPENING A MINE.

IN preceding numbers of the *Saturday Magazine**, a short and general outline has been given of the mode in which the mineral productions of Cornwall are obtained from the bowels of the earth, of the processes they undergo on the mine, and also a slight sketch of the progress of mining from remote ages. Resuming the subject at a somewhat greater length, we shall now proceed to describe the manner in which mineral veins are originally discovered, and to trace more fully the nature of the operations by which the miner is enabled to extract their metallic produce.

Familiar as every person must be with the varied uses and infinite importance of the metals in all the arts of life, information like the present can hardly fail to be interesting; and we may further remark, that there is no country in the world, of the same extent, which is known to possess the same degree of mineral wealth as our own. Indeed, if we except quicksilver and the precious metals, Great Britain may be said to contain, and generally, in considerable abundance, nearly all the mineral and metallic substances with which we are acquainted. It is to this liberal provision of nature that we may, without difficulty, trace much of that superior national importance, which so strikingly contrasts with the small territorial extent of the British empire.

In order to render the subject more intelligible, we must remind the reader that *veins* are tabular masses of mineral substance, which are frequently found occupying what were once doubtless vast *rents*, or *chasms*, in the rocky masses constituting the crust, or exterior, of the globe. These *veins* or *fissures* run for a very considerable extent in a horizontal direction, in which line they may sometimes be traced for several miles, and have, indeed, in no instance been followed to an actual termination. Their direction below the surface is generally more or less inclined from the perpendicular, and very commonly forms an angle of from seventy to eighty degrees with the horizon. The distance to which they extend downwards is unknown, for although at a certain depth veins generally become unproductive of the metallic ores, and are, therefore, no longer followed by the miner, they have never been known actually to terminate in this direction.

The width of veins is extremely variable, some being but a few inches wide, while others are many feet; indeed, the same vein is subject to considerable variations in this respect. Generally speaking, however, mineral veins do not in this country exceed three or four feet in width, although, in some few cases, they have been known to be as much as from twenty to thirty. When this occurs, it is commonly owing merely to a swelling or expansion of the vein, and does not continue for any very great distance in a horizontal direction. Mineral veins are also subject to many irregularities from the crossing, separation, and junction, of other veins, but which need not here be further noticed.

The most important circumstance, however, relative to mineral veins, is the nature of their contents, or the substances of which they are composed, and of which one or more of the metallic ores, usually forms a considerable proportion. The most abundant substance, however, which is found in mineral veins, is commonly a compact earthy substance, or *spar*, which is either crystallized, or what is termed by the mineralogist, *massive*, or without any definite form. This substance is generally termed the "veinstone," or in the north of England, the "rider" of the vein. The most common veinstones are quartz, fluor spar, calcareous spar, &c., and the ores are often, more or less, intermixed with the veinstone.

Now if we suppose a vein or fissure to be composed in great measure of either of these substances, the ore (which alone forms the object of the miner's research,) may be said generally to occur here and there in masses, of every possible variety of form and extent, with unproductive portions of veinstone interposed between them, the form and extent of which are, of course, equally irregular. It is from this circumstance, or the very irregular manner in which the masses of ore occur in veins, that most of the difficulties and uncertainty of mining arise, for although some general laws respecting their position may be recognised, they do not hold good in all cases, and the miner is therefore continually obliged to have recourse to actual trials.

It seldom happens that mineral veins are visible at the surface, as they are generally hidden by the vegetable mould and loose broken stone, which, in most cases, form a superficial covering to the solid rock wherein they are situated. Where, however, this covering has been swept away, as in the beds of rivers and torrents, or where the rock is laid open to view, as in precipices and cliffs on the sea-shore, they may occasionally be seen.

In this case, the first operation of the miner is to drive a horizontal passage, termed an "adit," *upon* the vein, following all its windings and irregularities. This passage, of course, he commences from the face of the rock, and at the greatest depth that may be convenient. In this manner the vein is laid open, and its contents exposed to view, and by excavating both above and below, the ore can readily be obtained, wherever it is sufficiently abundant. Should want of air or other circumstances render it desirable, a second outlet to the surface may be obtained by sinking a pit, or shaft, so as to communicate with the adit. In cases of this kind, the workings of the mine may proceed indefinitely, by continuing the adit upon the vein, and sinking shafts

* See Vol. III., p. 178, and Vol. IV. p. 43.

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as often as may be necessary. By means of these shafts, other passages or "levels" may be driven upon the vein, both above and below the first, if the ore is found to extend in these directions, and when thus laid open, may readily be taken away, or "worked out." This mode of opening mines will easily be understood, by reference to the annexed sketch.

When, however, the vein is not visible at the surface, it is generally discovered by digging trenches a few feet in depth, and crossing the direction in which it is probable veins may run, that direction being very generally, in this country, nearly east and west. These trenches are made of sufficient depth to penetrate the loose soil, and lay open the firm rock in which the vein is as it were imbedded. By this process, therefore, the miner is enabled to discover whatever veins may be situated in the tract he explores.

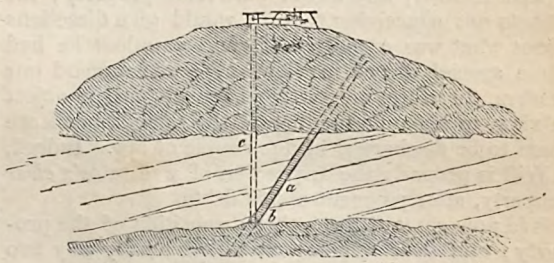
As it very generally happens, that the face of the country is not so mountainous and abrupt, as conveniently to admit of driving an adit in the manner before noticed, so as to explore the vein at a proper depth, mines are generally opened by sinking a shaft from the surface, in the manner shown in the annexed sketch. The direction and dip of the vein, or "lode," having first been ascertained, generally by sinking a few small pits for a short distance upon it, a spot is chosen for a shaft, determined by some promising indication the vein may exhibit there, or any other favourable circumstance. As mineral veins seldom, however, contain ore in any quantity, at a less depth than ten or twenty fathoms, the shaft is generally so placed as not to intersect the vein, till it reaches this or a greater depth. A rectangular space, usually about six feet by four, having been marked out on the surface, the sinking of the shaft commences, much in the same manner as that of a common well, the pick-axe and shovel being generally the only tools at first required. At the depth of fifteen or twenty feet, however, the vegetable mould and loose rubble, which always lie immediately under the surface, generally terminate, and the hard solid rock makes its appearance. Here then the work becomes much slower and more difficult, and the *pick*, and *gad*, the *borer*, and *mallet*, are put in requisition. If the rock is very hard indeed, the work is chiefly performed by the latter tools, the borer or *jumper* being driven into the rock, the hole charged with powder, and then blasted. When the ground is not so hard, the pick is often sufficient.

When the excavation has proceeded a very few feet below the surface, two very essential points must be provided for, the extraction of the stuff, and the support of the soil. A glance at the cut will show at once the manner in which this is accomplished. A very strong windlass, similar to that of a common well, is fixed over the shaft, and to each end of the rope a large iron bucket, or *kibble*, is attached, so that while the one is ascending, the other is descending, and while the upper one is being emptied of its contents at the surface, the lower one is being filled at the bottom of the shaft. In order to admit of sufficient power being applied, the windlass is provided with two handles, which are turned round by a couple of men, as shown in the drawing. The requisite support for the sides of the shaft is obtained, by placing within it, a timber framework, constructed in the following manner. At the mouth of the shaft, and at successive intervals of four or five feet below it, four pieces of strong timber are placed, framed together at the ends, and corresponding in form and size with the shaft itself. These horizontal timbers are further strengthened by uprights, connecting the corners, thus forming a continuous framework of great strength within the shaft. Small boards or *laths*, are then driven in, nearly close together, between the framing and the sides of the shaft, thus affording the necessary support all round. This kind of timbering is generally used only for the first twenty or thirty feet, as shown in the drawing, as the solid rock is generally firm enough to stand without support. Sometimes, however, shafts must be timbered nearly from top to bottom.

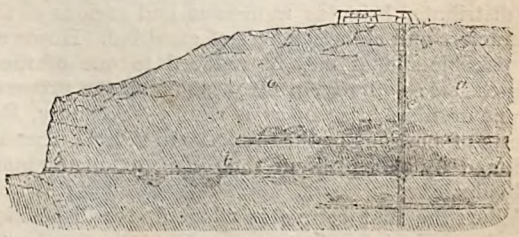
It is also necessary, even in this early stage of operations, to provide for ventilation, as the burning of the candles, blasting, and respiration, all tend greatly to vitiate the air, and the gases thus formed, being heavier than the atmosphere, will of course remain at the bottom of the excavation.

This evil is remedied by the following simple contrivance. A wooden pipe carefully rendered air-tight, is fixed along one corner of the shaft, reaching to within a foot or two of the bottom, and rising to seven or eight feet above the surface of the ground. At the top of this pipe a large square funnel is fixed, in such a manner that it can always be turned round, facing the wind. The air entering this funnel, having no other outlet, is obliged to descend to the bottom of the shaft, when rising again to the surface, it forces out the foul air before it, and thus a perfect ventilation is effected.

In the annexed drawing, the shaft has just attained its first object, that of *cutting* the vein. Two men are seen employed in sinking it, one is breaking the ground at the bottom with his pick, the other preparing to fill the descending bucket or kibble, with the ore and other substances which have been detached in forming the excavation. F. B.



MINERAL VEIN SEEN IN A CLIFF (*Cross view.*)



MODE OF WORKING THE VEIN. (*Side view.*)

a, the Vein. *b*, the Adit. *c*, the Shaft.

It was not till the time of Elizabeth that the English learnt the art of making needles, the manufacture of them had hitherto been carried on by foreigners.—*Life and Times of Henry the Eighth.*

He that makes light of small faults, is in a ready way to fall into great ones.

Sir Henry Sidney was the brave and virtuous father of the more renowned Sir Philip. To a friend of a fretful and quarrelsome temper, he said, "Take it from me, a weak man complains of others, an unfortunate man of himself: but a wise man complains neither of others nor of himself."

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ON WILLS. No. IV

Wills of Personal Property continued.

§ 6. *On the Form and Language of Wills.*

Having shown, in former papers, who may make a Will, and with what ceremonies a Will should be made, we now come to consider the form and language of the instrument.

These are mere questions of convenience; for a Will may be drawn in any form the testator chooses: he is not bound to any particular order of arrangement, nor obliged to use any set form of words: he may express his wishes in any manner he pleases, and the following hints are only suggestions of the best method of doing so.

A Will should begin with some such clause as the following;—"This is the last Will and Testament of me, A. B., of Chancery-lane, in the City of London, Grocer." Such an introduction is useful in removing all doubt about the nature of the instrument, or the identity of the testator.

If any directions are to be given about the place or mode of burial, these generally follow immediately after the introduction. No legal advice seems necessary on this topic.

The subject which usually comes next in order, is one which had better be omitted altogether. We mean the direction for payment of debts, and of the expenses of the Will and Funeral. This is a very superfluous clause; for the law will take good care that these debts and expenses shall be paid, and no executor is so ignorant as to need being reminded of his duty on this head. But it is worse than superfluous, as it sometimes raises doubts whether the testator did not intend his debts to be paid in a particular manner, and out of particular property; the Courts not conceiving that he would give directions about what was a matter of course, unless he had some special object in view. We recommend our readers, therefore, unless they have any special object in view, to make no mention of their debts, but leave them to be discharged in due course of law. Indeed, a Will is properly the disposition of a testator's *clear* property, after payment of his debts.

We now come then to this disposition of the property; and it is a point on which hardly any two wills can be alike; the modes in which property may be distributed being as numerous and various as the characters and circumstances of mankind. However, all possible modes may be reduced to one of these two classes:—Firstly, a Bequest of *all* the property to one or more persons as a whole.—Secondly, Bequests of portions of the property to several persons, followed by a bequest of the *residue* to one or more individuals.

When the first course is adopted, the disposition may be made in such terms as the following:—"I give and bequeath all my personal estate and effects, whatsoever and wheresoever, to C. D. for his own absolute use and benefit;" or "to C. D., E. F., and G. H., equally to be divided between them for their own absolute use and benefit respectively." The words "for his own absolute use and benefit" are not necessary, but may be useful to remove any suspicion that you intended C. D. to enjoy the property for his life only, or to hold it upon any trust. We shall speak hereafter of the mode of bequeathing property, so as to effect either of these intentions.

It is not uncommon for a testator, when disposing of all his property, to begin by naming certain particulars, and end the catalogue by words of a general description:—As, for instance, "I bequeath to A. B. all my stock in the funds, ready money, furniture, wine, plate, linen, china, and all other my property whatsoever." This, however, is not advisable, as it tends to raise a doubt whether, notwithstanding the general words at the end, the testator did not mean to confine his gift to property of the same nature only with the several articles mentioned. For instance, in the example we have given, it might be doubted whether the testator's leasehold property passed by the bequest.

When the second course of disposition is adopted, it is most natural to begin with bequeathing the several legacies, or portions of property, intended to be given, and to conclude with a bequest of the residue. This will lead us to consider the nature of Legacies.

§ 7. *On Legacies, and the Difference between specific and general Legacies.*

Any gift by Will is properly a Legacy, but the word is usually confined to gifts by Will of a *portion* of the testator's property. Now all bequests by a testator of a portion of his property, are either *specific* bequests or *general* bequests. A *specific* bequest, or legacy, is a gift of a specified portion of the property, distinguished from the rest. A *general* bequest, or legacy, is a gift of something which is to be paid, or satisfied, out of the general property of the testator, but which does not apply to one part of the property more than to another.

For example, if I bequeath "my gold watch," "the diamond ring, which was my mother's," "the 1000*l.* Consols now standing in my name," "the 50*l.* now owing to me from X. Y.;" these are *specific* bequests. On the contrary, if I bequeath "a watch worth 10*l.*," "a mourning ring," "a sum of 1000*l.* Consols," or "a legacy of 50*l.* sterling," these are *general* bequests. In the former cases, the legatee, (that is, the person to whom the bequest is made,) has a claim on a particular and specified portion of my property; in the latter cases, he has only a claim to have the gift made good in some manner out of my general property.

A specific legatee has an advantage over a general legatee in this; that, if the testator's property falls short of paying all the legacies in full, he, nevertheless, keeps the whole of his specific legacy, while all the general legacies are reduced in proportion. To compensate for this, he has a disadvantage; which is, that, if the specified portion of property bequeathed to him is lost or fails, he loses his legacy altogether, having no claim upon the general fund.

These results seem natural enough, and must generally agree pretty well with a testator's intention, when the subject of the specific bequest is a trinket, an article of furniture, a leasehold house, &c. But when the bequest is of money or stock, the law must often disappoint the wishes of testators, who, probably, neither intend their specific legatee to have a benefit at the expense of their general legatees, nor desire that he should suffer from an accidental failure of the specified fund. In bequeathing, therefore, money or stock, care should be taken not to make the bequest specific, unless the testator expressly wishes it to be so.

In bequests of money, it is not very easy to run into any error. A bequest of money is always a general bequest, unless expressed in terms which no man would be likely to adopt, who was not anxious that it should be specific; as a bequest of "the debt due to me from X. Y.," "the money in my iron chest," &c.

But, in bequests of stock, there is more chance of making a blunder; for the law-reports abound in

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fine distinctions as to what shall be considered a general, and what a specific legacy of stock. Ordinarily, a bequest of "*my* stock," or of "1000*l.* consols, part of *my* stock in that fund," is specific; while a mere bequest of "1000*l.* consols" is a general legacy. But this cannot be always relied on; and, perhaps, the safest plan is one we have seen adopted in some Wills, by which a testator bequeaths to A. B. "1000*l.* three per cent consolidated Bank Annuities *as a general, and not a specific legacy.*" W.

[To be continued.]

ADDRESS TO A GODCHILD.

We this week present to our readers an original paper, written by the late Mr. Coleridge about eleven days before his death, and addressed to a little child to whom he stood godfather a year or two ago. We do not remember ever to have perused a more affecting document. But our motive in requesting permission to publish this Address in the *Saturday Magazine* was connected with higher considerations. Mr. Coleridge had in the course of an eventful life, tried, and rejected, many of the prevailing errors of our religious sects. His deep convictions were not inherited, but obtained by patient thought, incessant labour, and fervent prayer for illumination. We here see the form of Christianity to which, on his death-bed, he set his seal. We believe there is not a sceptic in England who will venture to question the unique greatness of Mr. Coleridge's intellectual powers; and we are sure that there is no one who can, with a shadow of pretence, impeach his sincerity and entire disinterestedness.

To Adam Steinmetz K—.

My dear Godchild,

I offer up the same fervent prayer for you now, as I did kneeling before the altar, when you were baptized into Christ, and solemnly received as a living member of his spiritual body, the Church.

Years must pass before you will be able to read, with an understanding heart, what I now write. But I trust that the all-gracious God, the Father of our Lord Jesus Christ, the Father of Mercies, who, by his only-begotten Son, (all mercies in one sovereign mercy!) has redeemed you from the evil ground, and willed you to be born out of darkness, but into light—out of death, but into life—out of sin, but into righteousness, even into the "Lord our Righteousness;" I trust that He will graciously hear the prayers of your dear parents, and be with you as the spirit of health and growth in body and mind!

My dear Godchild!—You received from Christ's minister at the baptismal font, as your Christian name, the name of a most dear friend of your father's, and who was to me even as a son, the late Adam Steinmetz whose fervent aspiration, and ever-paramount aim, even from early youth, was to be a Christian in thought, word, and deed—in will, mind, and affections.

I too, your Godfather, have known what the enjoyments and advantages of this life are, and what the more refined pleasures which learning and intellectual power can bestow; and with all the experience that more than threescore years can give, I now, on the eve of my departure, declare to you, (and earnestly pray that you may hereafter live and act on the conviction,) that health is a great blessing,—competence obtained by honourable industry a great blessing,—and a great blessing it is to have kind, faithful, and loving friends and relatives; but that the greatest of all blessings, as it is the most ennobling of all privileges, is to be indeed a Christian. But I have been likewise, through a large portion of my later life, a sufferer, sorely afflicted with bodily pains, languors, and manifold infirmities; and, for the last three or four years, have, with few and brief intervals, been confined to a sick-room, and, at this moment, in great weakness and heaviness, write from a sick-bed, hopeless of a recovery, yet without prospect of a speedy removal; and I, thus on the very brink of the grave, solemnly bear witness to you, that the Almighty Redeemer, most gracious in his promises to them that truly seek him, is faithful to perform what he hath promised, and has preserved, under all my pains and infirmities, the inward peace that passeth all understanding, with the supporting assurance of a reconciled God, who will not withdraw his spirit from me in the conflict, and in his own time will deliver me from the Evil One!

O, my dear Godchild! eminently blessed are those who begin early to seek, fear, and love their God, trusting wholly in the righteousness and mediation of their Lord, Redeemer, Saviour, and everlasting High Priest, Jesus Christ!

O preserve this as a legacy and bequest from your unseen godfather and friend,

Grove, Highgate, S. T. Coleridge.
July 13, 1834.

He died on the 25th day of the same month.

THE TORTOISE, THE FROG, AND THE DUCK.

Along the fields one rainy day,
An aged Tortoise took his way:
His shell, like armour, on him leant
So heavily where'er he went,
That those who slightly looked at him
Had said he did not stir a limb;
But though his steps were short and few,
He had his walk, and liked it too.
Hop, skip, and jump! Now who goes there?
A speckled Frog, as light as air,
Deriding, as a piteous case,
The quiet creature's humble pace:
And lo, with empty folly tossed,
Full many a time his path he crossed;
Then stopping, panting, staring, said,
"You've got a house upon your head!
For if you were but fresh and free,
I'd bid you try a leap with me!"
Then head o'er heels the coxcomb rose,
Descending near his neighbour's nose.
"Boast not," the gentle Tortoise cried,
"The gifts that Goodness has supplied;
Nor seek, by conduct light and vain,
To cause less gifted creatures pain;
I, too, have blessings kindly lent,
And trust me, brother, I'm content;
My shell, for instance, like a roof,
Makes my old body weather-proof,
And guards me wheresoe'er I go,
From strong attack and secret foe."
"Why, as to weather," said the Frog,
I live in all, rain, sunshine, fog,
You've seen me dance along your path,
Now you shall see me take a bath!"
With that uprose the heartless fool,
Next moment splashing in the pool;
Quick moved his legs and arms; I ween
A better swimmer ne'er was seen:
Then on the bank the boaster sat;
"Now Tortoise! What d'ye think of that?"
A hungry Duck, who wished to sup,
Just at that moment waddled up,
And ere his sentence had its fill,
The Frog was quiv'ring in her bill!

O may I still contented be
With what kind heaven hath given me:
And though I do not *seem* so blest
As others, think my lot the best.
But more than all, I will refrain,
My lips from mockery and disdain. M.

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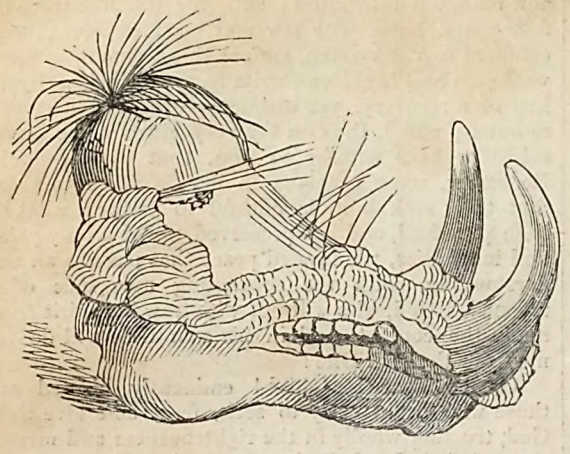
ON THE STRUCTURE OF ANIMALS. II.



HEAD OF THE ETHIOPIAN BOAR.

SKELETON OF THE WILD BOAR OF GERMANY.

The changes which we see in the forms of different animals, are referrible to one principle, the adaptation of the parts to their proper uses. We may, in some measure, consider the head in animals as performing the office of hands. The spine and the head, while they retain their offices of protecting the brain and the spinal marrow, and are permanent in regard to them, vary in their processes or shapes, and in their relations. Pursuing this idea, we shall be able to account for the characteristic forms of the larger quadrupeds.

The principle which will guide us here, as it will, indeed, in a more universal survey of animal nature, is that the organization varies with the condition or the circumstances in which the animals are placed, that they may feed and multiply. If we take into consideration any of the great functions on which life depends, we shall perceive that the apparatus, or the mode of action of the parts, is altered and adapted to every changing circumstance. Digestion, for example, is the same in all animals; but there is an interesting variety in the organization; and the stomach will vary in its form, and in the number of its cavities, according to the food received by the quadruped, or bird, or fish, or insect; a variation not depending upon the size or form of the animal, but adapted purely to the conversion of its particular food into nourishment. We shall find the gizzard in a fish, or in an insect, as perfect as in a fowl. So the decarbonization of the blood, is the same process in all living animals; but the mode in which respiration is performed, varies according to circumstances, and the apparatus is especially modified and adjusted to the atmosphere or to the water.

But although the organs subservient to the grand functions, the heart and blood-vessels, the lungs, the stomach, be variously adapted to the different classes of animals, they change much less than the parts by which animals are enabled to pursue their prey, or to obtain their food.

Their extremities, by which they walk, or run, or creep, or cling, must vary infinitely. And so their teeth and horns, and the position of their head and the strength of their neck, exhibit nearly as much variety as their extremities; because they, likewise, must be adapted to their different modes of obtaining food, or of combating their enemies.

When we look upon the boar's head, we comprehend something of his habits; and see what must be the direction of his strength. He feeds by digging up roots, and the instruments by which he does this, are also, those of his defence. The position of the tusk defends the eye in rushing through the underwood; and the formation of the spine, the remarkable ridge in which the back part of the scull rises, for the attachment of powerful muscles, all show the intention, that he shall drive onward with his whole weight and strength, so that he may rend with his tusks*. We now understand the reason of the shortness and inflexibility of the neck: because the power of the shoulders is directed to the head, and, we may say, to these large tusks. A long and flexible neck would have rendered these provisions useless.

What a complete contrast, then, there is between this animal, and any of the feline tribe; a contrast of form and motion at once referrible to their spine. In the tiger or leopard, we see the perfect flexibility of the body, and a motion of the spine almost vermicular, corresponding with the teeth and jaws, and the free motion of the paws.

* The sketch is from a dried head of the *Sus Ethiopicus*, with part of the scull exposed. The tusks show what a formidable animal it has been. That which rises out of the upper jaw is of great size, and we must admire the manner in which the tusk of the lower jaw closes upon that of the upper one, so as to strengthen it near its roots. The great size and sharpness of their tusks illustrate what is offered in the text, that the main strength of the animal must be directed towards them. The ring of the back of the head will be seen to correspond with the great height and strength of the spinous processes of the back, exhibited in the figure of the wild boar of Germany.

[Abridged from Bell's *Bridgewater Treatise.*]

Rare qualities may sometimes be prerogatives without being advantages; and though a needless ostentation of one's excellencies may be more glorious, a modest concealment of them is usually more safe; and an unseasonable disclosure of flashes of wit, may sometimes do a man no other service, than to direct his adversaries how they may do him a mischief.—Boyle.

Ambition breaks the ties of blood, and forgets the obligations of gratitude.—Sir W. Scott.

Mrs. Chapone was asked how it was she was always so early at church, "Because" said she, "it is part of my religion not to disturb the religion of others."

Sir William Gooch, governor of Williamsburg, walking along the street with a friend, returned the salute of a negro servant who was passing by; "I see," said his friend, "you condescend to take notice of a slave." "Yes" replied Sir William, "for I cannot allow even a slave to excel me in good manners."

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THE

Saturday Magazine.

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UNDER THE DIRECTION OF THE COMMITTEE OF GENERAL LITERATURE AND EDUCATION, APPOINTED BY THE SOCIETY FOR PROMOTING CHRISTIAN KNOWLEDGE.

STAFFA.

VOL. V.

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SKETCHES OF THE HIGHLANDS AND ISLANDS OF SCOTLAND.

PART THE SECOND.

IONA; INSULAR CHURCHES.

EARLY on the 25th, (A.D. 1827,) we started for Staffa and Iona. Partial gleams of sunshine illuminated the bold, rugged head-land of Ardnamurchan, and were reflected dimly from the distant, lofty, and conical summits of the isle of Rum. The point of Calliach, in Mull, was sheathed in foam, by the waves of a wild sea, mingling their hoarse uproar with the shrill cries of innumerable sea-fowl, hovering around its summit. It is said that the poet Campbell, who resided for some time in the vicinity of this promontory, often selected it as the scene of his lofty musings, as he listened to the roar of the distant Corry-vreckan, that it was the birth-place of the *Exile of Erin*, and of much of the *Pleasures of Hope*. The grouping of the numerous islands, off Mull, is extremely picturesque: Staffa, amongst them, rearing its basaltic pillars, forming a long causeway, gradually terminating in a majestic colonnade, crowned by a green and overhanging brow. The southern front encloses the caves within its triple buttresses.

Before us, Iona reared its lonely tower from the bosom of the stormy deep. The island contains 350 inhabitants, part of whom are congregated in a village near the church. The celebrated ruins consist of a cathedral, a nunnery, and St. Oran's chapel. The cathedral is small and cruciform: the height of the tower is only 70 feet; its architecture is rude and inelegant. On the north side of the altar is the tomb of Abbot Mackinnon, who died in A.D. 1500, and is represented in a recumbent position. Of the nunnery, the remains are scanty. St. Oran's chapel contains some tombs, and is surrounded by the principal remaining monuments, unfortunately, much defaced by weather and the footsteps of visiters. In this hallowed cemetery, this conventional asylum of the dead, which religion or superstition happily respected, even amid the fury of perpetual warfare, repose, according to dubious tradition, the bones of upwards of forty Scottish, besides French, Irish, and Norwegian kings; and of many lords of the isles, bishops, abbots, and chieftains; some of whom are represented in full armour, cross-legged, with their hunting-dogs at their feet; and some, the Macdonalds, a clan of Norwegian origin, indicated by their appropriate armorial bearing, the warlike galley.

An instance of the verification of tradition, no less than of this practice of burying their favourite animals, the companions of their sports, with the dead, was mentioned to me by the minister of Glenelg, who was present at the opening of one of two mounds, in that parish, on the shore of the Sound, opposite to Sky, supposed to contain the remains of two giants who could leap across the Sound, buried together with those of their hunting dogs. At some depth was found a large flat stone, covering many small bones, which proved, on examination, to be those of dogs, deposited on a bedding consisting of several layers of earth, covering a flat black stone, resting upon a stratum of the finest gravel, resembling gold-dust, four inches in thickness; and underneath, emerged two large bones, one of the jaw, the teeth of which were perfect, and the other of the arm. A boy, struck with the size of the jaw-bone, clapped it forthwith on the chaps of the fattest man in the parish, which it fairly encased. Awe had mingled much with the curiosity which had tempted these people to violate the sanctuary of the dead; and a thunder-storm happening, as is invariably stated to be the case on such occasions, at the moment of the sacrilegious joke, preserved the other tomb from similar profanation*.

The present ruined structures of Iona were erected by their Romish possessors, who held them till the Reformation, at which period they did not escape the destroying edicts of the Presbyterian synods.

The destruction of churches, and the spoliation of church property, at the period of the Reformation in Scotland, were carried to a great extent; and the Highlands and Islands have been, consequently, very inadequately supplied with spiritual instruction. Dr. Macculloch calculates, on the collective authority of old writers, that the Western Isles alone contained no less than 300 churches and chapels in the Romish age. Of these, many were merely cells, or occasional stations for divine service, and to many of them, probably, no priest was appointed. Their erection must be attributed to the piety or the superstition of the chiefs and people, no less than to the influence of the priesthood.

These remnants of Romish zeal occupy the site of more ancient monuments of the apostolic piety and zeal of their founder St. Columba, who came to Britain, to preach the Gospel to the northern Picts, in the year 565. This eminent missionary held Christianity, uncorrupted by many errors which subsequently infected it, and instituted the order of the Culdees, who maintained their ground in various parts of Scotland, against the usurpations of the Romish See till the fourteenth century, when they were suppressed. St. Columba was superseded by St. Andrew, as patron saint of Scotland, on the transfer of the primacy from Dunkeld to St. Andrew's.

Stripped of all that is fabulous or uncertain, the real history of Iona, a sanctuary erected in a dark age, on the lonely beach of a remote island, amid tribes of pirates and freebooters, must inspire a solemn and grateful recognition of the peaceful triumphs of the Gospel, and of the overruling influence of Divine Providence, in employing even the superstition of mankind in protecting and perpetuating its ascendancy, till purified by reformation, it shines forth, amid surrounding gloom, in its pristine light and lustre.

In Johnson's powerful and acute understanding, the caustic shrewdness of the critic too often prevailed over his poetical feelings; yet of his susceptibility to the poetry, no less than to the charity of that religion which he loved and venerated, as essential to the peace, the dignity, and happiness of mankind, the immortal passage which records the emotions excited in his breast, by the prospect of Iona, affords unquestionable proof.

"We were now treading that illustrious island, which was once the luminary of the Caledonian regions, whence savage clans and roving barbarians derived the benefits of knowledge, and the blessings of religion. To abstract the mind from all local emotion would be impossible, if it were endeavoured, and would be foolish, if it were possible. Whatever withdraws us from the power of our senses; whatever makes the past, the distant, or the future, predominate over the present, advances us in the dignity of thinking beings. Far from me, and from my friends, be such frigid philosophy, as may conduct us, indifferent and unmoved, over any ground which has been dignified by wisdom, bravery, or virtue. That man is little to be envied, whose patriotism would not gain force upon the plains of Marathon, or whose piety would not grow warm among the ruins of Iona."

Yet it was not till the present day, that the inhabitants of Iona fully enjoyed the benefits of the Gospel. When a modern missionary, Legh Richmond, visited the island, Divine service was performed in it only four times in the year. Mr. Richmond repeatedly addressed these islanders: and his account of his visit to them, is one of the most interesting productions of his well-known pen. A parliamentary church has been since erected, and a minister appointed to the island.

Religion and civilization usually keep pace together. The people of Iona, almost unprovided with religious instruction since the period of the Reformation, were pointed out as in the lowest stage of improvement. It is one of the islands in which the old-fashioned hand-mill, called the quern, was last used; a pair of stones, rubbed by two persons, identical with that employed in Eastern countries, according to the Scriptural description: "Two women shall be grinding at the mill," and, doubtless, derived by the Hebrideans from their Eastern ancestors.

Legh Richmond says, when desirous of rewarding the children of the school with a public entertainment, that "the best sheep to be found in the island, was purchased for the vast sum of six-shillings. But a difficulty arose on the occasion; there was fuel to roast the creature, but the whole domain could not supply the necessary apparatus for its dissection. The children assembled on the shore,

* The practice alluded to, reminds us of the superstition of the American Indian, described by the poet:

"Who thinks, admitted to an equal sky,
His faithful dog shall bear him company."

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and picked up shells, to answer the purpose of knives and forks."

Of the want of surgical assistance, we witnessed a melancholy proof. The schoolmaster, our guide, anxiously inquired of us when we landed, whether a professional man was of our party, and soon conducted us to a cottage, where a poor fisherman displayed an arm dreadfully swollen, in consequence of the prick of the fin of a gurnet; the scrofulous and inflammatory state of the blood of the Hebrideans produced by their scanty or fish diet, not unfrequently occasioning death upon the slightest injuries. This poor fellow suffered subsequently amputation of the finger, but whether his life was preserved, we did not hear. The party from the steam-vessel furnished medicines, and subscribed to procure the assistance of a surgeon from Tobermory.

Nor has learning lagged behind, in investigating the antiquities and preserving the ruins of Iona. An *Iona Club* has since (1833) held its first meeting on the island, under the auspices of the Duke of Argyle, the proprietor, and the principal of the University of Edinburgh, who himself attended the meeting.

STAFFA.

On returning from Iona, we landed at a small cave on the eastern shore of Staffa, where the basaltic causeway commences. It is very broad, of unequal height, sloping like a glacis from the water to the base of the higher columns. The dimensions of the caves it will be unnecessary to detail. The largest can be explored in fine weather by a boat. The sea was now boisterous, and we could penetrate into its recesses only by means of a narrow and broken ledge, which raised us above the water. And in the interval of the waves which burst in, rushing to the farthest extremity, deafening us with their uproar, and hurling aloft a volume of spray, and flakes of foam, we could admire, in mute astonishment, the beautiful symmetry of this wondrous pile, resembling, yet surpassing the imitative efforts of man: the regular arrangement of its massy columns; the richness and variety of the tints which adorn them, more brilliant than the hues which the painted panes of the window of a Gothic church shower on its clustered pillars; the dark shadows afforded by the intermediate recesses; the sombre grandeur of the ponderous roof, and the smooth pavement which the sea supplies, when tranquil, to the stately temple. To borrow the language of the poet:—

> Here as to shame the temples deck'd
> By skill of earthly architect,
> Nature herself, it seem'd, would raise
> A minster to her Maker's praise!
> Not for a meaner use ascend
> Her columns, or her arches bend;
> Nor of a theme less solemn tells
> That mighty surge that ebbs and swells,
> And still, between each awful pause,
> From the high vault an answer draws,
> In varied tone prolong'd on high,
> That mocks the organ's melody.
> Nor does its entrance front in vain
> To old Iona's holy fane,
> That Nature's voice might seem to say
> "Well hast thou done, frail child of clay!
> Thy humble powers that stately shrine,
> Task'd high and hard—but witness mine!"
>
> *Lord of the Isles.*

And most fully did the different emotions evinced by the visiters at the prospect of the one and of the other of these kindred structures justify this vaunt, poetically imputed to Nature. The party who had run riot over the ruins of Iona exulting in loud mirth over the credulity of the cicerone, were now fairly awed into silent wonder by indications of the presence of superior power which no scepticism could dispute. One aged man of our number, of clerical appearance, having doffed his hat, stood resting against a pillar, whilst his uplifted eyes and the motion of his lips indicated the feelings which, more or less, actuated all present.

MULL.

On our return from Staffa, a boat pushed out from the Bay of Quinish, in Mull, and soon introduced us to the hospitality of a neighbouring mansion. The road hence to Tobermory, was said to be the only one in this large island; some difference among the proprietors preventing the formation of the Parliamentary roads, which have extended to the farthest limit of the remoter island of Sky. A lady living on the south-east side of the island, when she visits her distant friends, is conveyed by eight Highlanders, on a litter.

Ballachroy affords a pleasing specimen of a Highland village: it consists of a single street, composed of neat but small houses: a good school-house, a public-house, and the church rising on an adjacent eminence. To each house is attached a small piece of ground, called a *croft* on which vegetables are raised for the use of the family. and hence the tenant is called a *crofter*. The numerous little patches in the neighbourhood of the village, belonging to these people, and the herd of cows, each keeping one or two, ranging the adjoining hill under the charge of a single lad, awaken the pleasing conviction of independence and comfort. Happily for England, its landed proprietors are now gradually adopting the excellent plan of allowing small allotments of land to their cottagers, the benefits of which will prove incalculable.

On the desolate brow of a steep ridge beyond Ballachroy, we passed a small hamlet, consisting of a dozen cottages, called *Siberia*. On asking one of the inhabitants wherefore they chose so dreary a situation for their abode, he replied, looking archly, "Coll knows the reason very well." This was, in fact, the place to which the Laird of Coll banished his people from that Island, when guilty of smuggling, and other crimes deserving exile. They had no alternative but to accept this punishment, or to fly the country: a proof of the practical continuance, under the control of law, of the *heritable jurisdictions* of the chiefs, which were abolished by Act of Parliament in 1748. Dr. Johnson adverts to it. "Many of the smaller islands have no legal officers within them. I once asked, if a crime should be committed, by what authority the offender could be seized, and was told that the Laird would exert his right; a right which he must now usurp, but which merely necessity must vindicate, and which is, therefore, yet exercised in lower degrees by some of the proprietors, when legal process cannot be obtained."

The present criminal jurisdiction of Scotland (1827), is exercised in the first instance by the sheriff-depute*, or deputy-sheriff, in each county, a salaried officer, holding of the king as sheriff of the kingdom, and by his delegates, the sheriff's-substitutes; some of whom, usually tacksmen, or persons holding farms on lease, are appointed in the islands. In some of the islands, as for instance in Jura and Colonsay, the proprietor of the island acts, in virtue of a commission from the lord lieutenant, as magistrate.

Dr. Johnson remarks, that "where the chiefs were men of knowledge and virtue, the convenience of a domestic judicature was great." That the present Laird of Coll has exemplified, in his administration of local justice, the truth of this observation, may be fully inferred from the testimony still borne to his conduct, by the numbers who still resort to him to submit their differences to his arbitration. He is younger brother of the amiable companion of Dr. Johnson, who was drowned on the coast of Mull ere the pages commemorating his virtues were committed to the press. The influence of a resident landlord in these regions, when steadily and beneficially exerted, is unbounded. Independently of the power which he enjoys as lord of the soil, without rivalry or counteraction, he is regarded with a portion of that homage which the chiefs, in times not beyond the memory of living man, received from their clansmen.

We passed the head of Loch Frisa, and viewed from its shore the lofty summit of Ben More. Mull is, with the exception of some patches of arable land, a vast moor. Near Tobermory is a sequestered scene, of much beauty, recalling to the Italian traveller the recollection of Terni. Sacheverel, 150 years ago, was struck with its resemblance to Italian scenery. A lake is enclosed by an amphitheatre of hills, covered with oak, interspersed with torrents, forming picturesque cascades. On its bank stands Coll castle, a commodious mansion, erected by the son of Mr. Maclean, on land once confiscated, but repurchased. The Macleans have suffered by every quarrel since the conquest of Cromwell.

On Sunday we attended Divine service, in the parish church of Ballachroy. The dress and general appearance of the congregation were highly respectable. The minister officiated in the Gaelic language; but another, who was of our party, supplied afterwards a service in the English; and the congregation, though the greater part understood not the language, remained till its conclusion. English

* He is now called sheriff.

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RUINS OF IONA.

travellers are frequently indebted to the courtesy and kindness of the ministers of the Scottish kirk, in performing an additional service almost exclusively for their benefit.

The large parish in which Ballachroy is situated, contains four preaching stations, one of which is the above-mentioned kirk. A preaching station by no means implies a church or chapel: at the head of Loch Lomond, the only structure for the celebration of Divine service is an enormous stone. At Tobermory a barn is employed for the purpose; but a parliamentary church will be erected here. This parish is far better supplied with ministers than many in these regions. Besides the incumbent there are two *missionaries*. The office, and even the designation of *curate*, is offensive to the Scottish church; and in order to combine the adequate provision of spiritual superintendence with ecclesiastical discipline, when the extent of a parish requires more than one pastor, a portion of it is detached and confided to the care of a missionary, who is wholly independent of the parochial minister, but enjoys no corporate jurisdiction, being inadmissible to the presbytery, or other ecclesiastic courts, though subject to its authority*. Of the missionaries thus employed, twenty-eight depend on the commissioners of the General Assembly for managing the Royal Bounty, and eight on the Society for propagating Christian Knowledge in Edinburgh. The number still falls far short of that which the country requires. And there is scarcely a parish in the Highlands or Western Islands of Scotland, part of which, and, in many instances, the greater part, is not virtually extra-parochial. The combined efforts of societies and of government; the legislative augmentation of the incomes of the clergy of the Scottish church, to £150 per annum, when not amounting to that sum; and the parliamentary grant for the erection of the forty churches and manses, have very imperfectly supplied the great deficiency in the provision of the Church Establishment, impoverished at the Reformation. Previously to this period, Mull was divided into *seven* parishes: their number is now reduced to *three*. Nor has the unoccupied ground been cultivated by Dissenting Sects: the people are too poor to support their ministers. The *Seceders*, whose numbers amount to a third of the population in the Lowlands, and wealthier parts of Scotland, furnish only three ministers who preach Gaelic, and have not a single place of worship in the large shire of Sutherland. There are some Baptists, Independents, and Haldanites in the islands; and in some parts the Catholics predominate. But of these different sects hereafter. It is sufficient to observe, that the visionary, unphilosophical, and erroneous notion of those who imagine that the ordinances of religion can be generally maintained in a country without a Church Establishment, may be brought to the test even within the precincts of the British Isles. This dangerous opinion is an inference from the success of ministers depending on the contributions of their flocks, in wealthy towns or districts; and is overthrown by an extension of the view to those rural and poor parts of the country, which, whilst dwelling in the hot-beds of spiritual instruction, we are apt to overlook.

Inadequate as the Church Establishment is, in these regions, it has preserved a large portion of the population from the darkness of ignorance, or the false light of superstition.

* The ecclesiastical jurisdiction in Scotland is vested in the presbytery, composed of the ministers and some of the elders of a few adjoining parishes: in a synod, which comprises several presbyteries, and is composed of ministers and elders, elected by the presbyteries: and of the General Assembly, similarly constituted, in which the ultimate appellative jurisdiction is vested.

ARISAIG.—HOSPITALITY.—KELP.—ROMAN CATHOLICS.—EDUCATION.

On the 30th, we made an excursion in a yacht; an excellent cutter of thirty tons, built in Coll, of materials furnished by the island, *i. e.* by the wrecks of vessels. A shoal of whales, and a wreck, contribute no little to the slender stock of insular wealth. It is not insinuated that the islanders are, in the ordinary sense of the word, *wreckers:* they invariably resist the imputation, and transfer it to their more northern neighbours:—Orkney and Shetland serving as scapegoats to the Western Isles. Excepting some rare instances of criminal neglect towards those whom misfortune has cast upon their coasts, the natives of all these islands have treated them with hospitality; and the charge would be more satisfactorily repelled by a general denial, than by a vindication of themselves at the expense of their neighbours.

We passed the formidable front of Ardnamurchan, and the ruined castle, Tirim, a strong-hold of the Macdonalds, in the Bay of Moidart, shaping our course to Arisaig. On our left lay the *small isles*,—Muck, low and flat; Egg, rearing its Scuir, a lofty porphyritic tower; Rum and Canna in perspective; and the Coolin hills of Sky bounding the horizon. Two whales gratified us by their gambols: one closely chasing a shoal of fish, heaving his back above the water, and pouring forth from his nostrils a fountain of brine, around which swarmed flocks of sea-fowl, employed in the pursuit of the same game.

The proper season for witnessing the splendid exhibition of sea-fowl which the northern coasts of Scotland afford, is June, when rock, air, and water, teem with myriads of those delighted beings, glittering in sun-shine,

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or gemming the clouds; and the ear is deafened by the loud and discordant chorus in which they celebrate, unconsciously but unequivocally, the beneficence of their Creator.

The Bay of Arisaig is deep, but its entrance is narrow, and obstructed by rocks and islets. We landed for a legal purpose. One of our party was a writer to the signet and notary public, and proceeded to institute two *infeoffments*, i. e. to take possession *virtually* of the fee simple of the estate of a laird. The ceremony was simple, indicating its origin in ruder times, and partaking, in this respect, of the general character of the laws of Scotland relating to property, which are strictly feudal. The officer in question took his station on a small knoll: one of the crew of the cutter represented the attorney of the person infeoft, and received from the hands of the other, who acted as baillie for the proprietor, a tuft of grass and a shilling, repeating after the notary public the words of the usual formulary. The transaction required two witnesses, whom our party furnished, and who signed their names to each page of the instrument, which narrated what had taken place.

Rum was our next point; and we had rounded Egg, and already beheld the magnificent precipices of that mountainous island, overhanging us in sombre majesty, when the wind, which had been gathering, burst from all its peaks and gullies, and, damaging our rigging, drove us back on Egg, and restored us, about dusk, to our anchoring-place in the Bay of Arisaig. We soon espied a boat pushing forth gallantly towards us; and one of the rowers, evidently above the rank of a fisherman, invited us to come ashore, promising us hearty hospitality; and, on landing, we perceived, indistinctly, the form of a person wrapped in a Highland plaid, who acknowledged our approach by a slight inclination of the head, and led us through a ravine to a house about a mile distant, where we were introduced, dripping with wet, into a room, illuminated by a blazing fire, containing a large table covered with tea and divers viands, and surrounded by a family who welcomed us with genuine hospitality, as expected guests to bed and board. The sudden and unexpected transition from a stormy sea to comfort, good cheer, and cordial welcome, enabled us to realize, in some degree, the truth of Dr. Johnson's observations on the origin of romantic feelings suggested by similar circumstances. The proprietor had notice of our visit to the coast in the morning, and, from the promontory above his house, traced our vessel till the necessity of its return became obvious; when he immediately made ready for our reception. He and his ancestors had occupied this house during 200 years. Wind-bound, we enjoyed his kindness during three tempestuous days.

The Bay of Arisaig is a favorite resort of seals: they are constantly seen basking on the island, and scuttle into the sea when alarmed. The seal often follows a boat, rearing its head, resembling that of a bull-dog, but more hideous, above the water. The Highlanders attribute this habit of the animal to its love of music.

> Rude Heiskar's seal, thro' surges dark,
> Will long pursue the minstrel's bark.
> *Lord of the Isles.*

But we had no music for the entertainment of our pursuers but that of a rifle, which was successful in one instance. The animal shot sunk, and floated afterwards ashore. The natives attribute their frequent disappointment, in not securing the wounded seal, to his determination to deprive his destroyer of his prize; securing himself to the bottom of the sea by a network of sea-weed. The price of the skin varies from twelve to twenty-four shillings. The public-house, at the head of the bay, contains a large assortment, for sale, of the skins of seals, wild-cats, pole-cats, and otters. The wild-cats are very numerous and destructive in Scotland; sometimes growing to double the size of the common cat, and invariably of grey colour.

The cottage of Arisaig, the residence of Clanronald, chief of one of the clans of Macdonald, is situated about a mile from the head of the bay, commanding a pleasing prospect of a broad valley, laid out in plantations or pasture; surrounding a small and picturesque loch, enclosed by a range of copse-covered hills, embracing the bay in its semi-circular sweep. Lord Macdonald is chief of another branch of the Macdonalds; and the precedence of these two representatives of potentates, once exercising almost regal sway, is matter of dispute among genealogists. Sir Walter Scott assigns it to Clanronald.

Clanronald succeeded to an estate comprehending a territory on the mainland, bordering on the *country* of Macdonell of Glengarry, the isles of Egg, Muck, and Canna, together with South Uist and Benbecula, in the Long Island; but much of it was now on sale.

The maritime proprietors in Scotland have suffered material loss from the diminution of the price of kelp. This alkaline sea-weed, the use of which in the manufacture of plate-glass and soap, was unknown till the commencement of the last century, has proved, till within a few years, a fertile source of wealth. Some proprietors have raised as much as 1300 tons in a single year. The price of the ton, formerly amounting to 15*l.* and 20*l.*, has declined to 4*l.* and 5*l.*: and, except in some few instances the manufacture of kelp is unprofitable. The reduction of the duty on Spanish barilla, and the subsequent discovery of an artificial salt, have provided the market with a cheaper substitute. Of the tenures in the remote island of Scotland adverted to, of the produce of the kelp, and of the burden which the support of the population employed in it has entailed on the proprietor, the following extract from the evidence of Mr. Hunter, published in the *Third Report of the Emigration Committee*, presents a brief account. "The islands of South Uist and Benbecula, contain a population of about 6000. There are 489 small tenants or crofters, who pay rents from 1*l.* to 21*l.*, averaging 6*l.* 17*s.* 4*d.*; fourteen large tenants, who pay rents from 32*l.* to 400*l.*: there is one more who pays 400*l.*: these average 86*l.* 15*s.* Under these fourteen large tenants, there are 207 sub-tenants. There are annually manufactured about 1200 tons of kelp on Clanronald's estate at Uist. The kelp does not belong to the tenants, as in the Duke of Argyle's case, for the manufacturing of which, they receive from 50*s.* to 60*s.* per ton, which as nearly as possible discharges their rent. On this estate, about one-third of the population possess no lands. To keep these people alive, Clanronald expended, in 1812, 3353*l.* 7*s.* in purchasing meal for their consumption; in 1815, 111*l.* 11*s.* 3*d.*; in 1816, 242*l.* 8*s.* 3*d.*; in 1817, 4567*l.*; in 1818, 1136*l.* 19*s.* 8*d.* The kelp belonged to him as proprietor, but there was a deficiency of rental to the extent of these grants."

It must be observed, that the small tenants, or crofters, are usually tenants at will: but the large farmers in general hold leases of nineteen years. The farms in this neighbourhood, have been much transmitted in hereditary descent: the business of the country and of the police is conducted by the higher tacksmen and farmers.

The numbers stated to be thrown out of employment by the failure of the kelp-manufacture, in a memorial prepared at Edinburgh, in the beginning of 1828, by the proprietors of the western maritime estates, amounted to 50,000. The disposal of the superfluous population will be considered, when the question of emigration from these regions engages more particularly our attention.

Near the hamlet of Arisaig, stands a neat church between two well-built houses, which might be fairly presumed to be the manse and the school-house. And such they are: the former occupied by a Protestant *missionary*. The church, however, proved to be a Roman Catholic chapel. The proprietor, who erected it, offered the lease to the kirk in the first instance, and, on its being declined, to the present occupant. On a neighbouring hill remain the four walls of the ruined parish-church. A Catholic gentleman of our party referred its demolition to the era "when John Knox set about harrying the rooks." This act, contrasted with the immense predominance of the Catholics, who compose 19-20ths of the entire population in this neighbourhood, brings to mind the well-known story of the old woman, who set fire to her house for the purpose of destroying the rats: the house was burned; but the rats escaped. It must be recollected, in justification of Knox, that the destruction of churches, though attributable in part to the violence of his discourses, was contrary to his wishes, and that he struggled perpetually to secure the adequate endowment of the church, in opposition to the cupidity of the lay reformers.

The state of the Roman Catholics in the Highlands and Islands of Scotland, is a subject of much interest.

The natives of these regions were swayed by other impulses than those which affected the kingdom of Scotland in general, in which, at the period of the Reformation, the Catholic clergy were simultaneously deserted by their flocks. And the Romish religion is still professed by a considerable portion of them, intermingled with the Protestants, in a singularly chequered manner. They abound principally in Inverness-shire: constituting in Arisaig, which may be considered the central station of

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their faith, in Moidart, Knoydart, and the neighbouring island of Egg, 5-6ths of the population; in Glenelg, beyond Knoydart, about one-half: diminishing in the districts of Kintail, Loch Alsh, and Applecross, in Ross-shire, till they disappear on the borders of Cromarty and Sutherlandshire. They abound in Glengarry, in the country of Lovat; and further eastward, are still numerous on the property of the Duke of Gordon, which lies in the shires of Murray and Banff.

In the islands of Rum and Muck, on either side of Egg, Protestantism is universal; in Canna, popery predominates; in the large island of Sky, Protestantism exclusively. In the Long Island, popery prevails in Barra, Benbecula, and South Uist, and has extended partially into North Uist; whilst in Harris and Lewis, the northern islands of this chain, the people are all Protestants. The Southern Hebrides and Highlands contain very few Catholics.

The Catholics have, hitherto, maintained two establishments, one in Aberdeenshire, and one in the island of Lismore, at the entrance of the Caledonian canal. But the want of funds has rendered consolidation necessary, involving the discontinuance of that at Lismore.

This remarkable distribution of the two sects may be attributed to the original imperfect progress of the Reformation in these regions; to the inadequacy of the Protestant establishment and of education, and to the prompt submission of the people to the authority of their chiefs. Dr. Macculloch considers the influence of their religious feelings on the minds of the Highlanders, too powerful to admit of the operation of the last mentioned of these causes; and Mr. Glassford, in his letter to Lord Roden on Irish education, naturally enough attributes the effect to the insufficient supply of Protestant instruction. But the ascertained facts warrant the supposition of the influence of more than one cause.

The vast estates of Clanronald and Lord Macdonald, continental or insular, are contiguous. Yet the former are peopled almost exclusively by Catholics; the latter by Protestants. The grandfather of the present Clanronald was a Catholic. Lewis and Harris, and part of Sky, which were, and are still, partly the property of the Macleods, contain only Protestants. May it not be inferred that the authority of the chiefs, as well as the spirit of clanship, contributed to this result? The insulated existence of the Catholic religion upon the property of the Duke of Gordon, notoriously resulted from the long-continued adherence of that family to the Romish faith. The equally anomalous fact, that the religion of Rum should be Protestant, whilst that of the adjacent islands of Egg and Canna should be chiefly Catholic, may be explained by direct reference to the exercise of authority.

That the long neglect of the spiritual instruction of the natives of these remote regions, has contributed also, much to the preservation, and indeed, the extension of popery, is doubtless true. It accounts for the conversion of the people of Barra to that religion, who were all Protestants till after the Restoration; when upon the establishment of the Church of England in Ireland, some Irish priests fled from that kingdom to this island: and as Harris and Barra formed one parish, and the minister resided in the former, at too great a distance from the latter, availed themselves of the ignorance of the natives to effect their entire conversion*. The more recent progress of popery in North Uist, there being only four Catholics in that island at the period of the survey, may be traced to the same cause.

The supposition of the universal diffusion of education in Scotland is a common mistake, and affords a striking proof of the ignorance of the real state of the highlands and islands of Scotland, prevalent in England, and, indeed, till within few years, in the Scottish capital and Lowlands.

The *parochial* system of instruction which has obtained well-merited celebrity, and has been found so efficacious in the Lowland parishes, was of little avail in those vast regions, in which the parishes often extend from thirty to sixty miles, and consequently the parish schools are inaccessible to the greater part of the inhabitants, and, if within their reach, useless to a population acquainted only with the Gaelic language, which is not taught in those seminaries. The schools established by the Society for Promoting Christian Knowledge in Scotland, which conferred much benefit on the more neglected parts of the country, were restricted by a similar limitation.

In proof of the ignorance necessarily resulting from the education being exclusively in a language which the people did not understand, the recent returns published by the Inverness-shire Society, exhibited a ratio of *thirty to twelve in the hundred*, uneducated in the Orkneys, and Shetland Isles, and in those parts of the North of Scotland, where English is spoken generally, and as *seventy to the hundred*, in the western Islands and Highlands. To remedy this great deficiency, various societies were formed at Edinburgh, Glasgow, and Inverness. The Gaelic School Society of Edinburgh, instituted in 1811, adopted the opposite system of teaching only the Gaelic language, formed on the obvious principle which, so long the subject of debate and apprehension, may be now considered an axiom of education as of common sense, that a person can receive religious instruction, only in a language which he understands. It had been previously applied successfully to the education of the Welsh and Irish. The object of the institution is, the formation of circulating schools, or schools, which after having been fixed in one district for a period, varying according to local exigencies, are removed to another, and thus diffuse the benefits of the society, as generally and impartially as possible. The scholars pay no fees, and are liable to no expense, but that of contributing in turn, a portion of the fuel consumed in the school. The house is built by subscription of the neighbouring gentry, farmers and others. The schoolmasters are examined as to their piety and general qualifications, by the Edinburgh Committee, and are precluded from transgressing the limits of their prescribed duty, by an express regulation, which prohibits the teachers being preachers, or public exhorters, stated or occasional, of any denomination whatever. The Holy Scriptures are exclusively the subject of instruction.

The Society directed its first efforts to the supply of the religious wants of the people, of whom by far the greatest part of those of the Highlands and Islands can seldom, and some never "hear sermon:" to use the expression by which attendance at public worship is commonly designated, notwithstanding the appointment of Missionaries in some parishes; and with this view confined itself to teaching Gaelic as the only general vehicle of instruction, and to the use of the Scriptures, common both to Catholics and Protestants. They justly considered a more comprehensive system of education of secondary importance, while tens of thousands were ignorant of the simple rudiments of Christianity. Each report of this society affords a list, not only of the schools, and number of scholars actually in attendance, but also, all the places at which schoolmasters were formerly stationed; thus presenting a complete view of its operations, since the formation of the Society.

The number of children which received instruction from the Society in 1827, amounted to between 4 and 5000. But still, the efforts of this and other similar Societies, left a great void unoccupied; and the General Assembly of the Kirk of Scotland had become so sensible of the deficiency, that they appointed, in 1824, at the proposition of Dr. Baird, Principal of the University of Edinburgh, a committee for the purpose of ascertaining its actual extent, and of devising a plan for remedying it. Queries were sent to the ministers of the different parishes, and answers, the result of personal investigation, elicited; and the mass of valuable information thus collected is comprised in a series of voluminous Reports, containing probably the most accurate and complete representation of the deficiency of schools throughout Scotland.

The following brief statement, extracted from the Society's first Report, and confirmed by subsequent inquiries, presents a summary view of the state of education in different synods, pointing out clearly where the deficiencies principally exist.

"The whole population of Scotland amounts to 2,093,856, and the Church is divided into sixteen synods. In the ten synods of Lothian and Tweeddale, Merse and Tiviotdale, Dumfries, Galloway, Glasgow and Ayr, Perth and Stirling, Fife, Angus and Mearns, Aberdeen, and Moray, there are 764 parishes, and 1,716,126 persons; and so abundant is the number of schools in these districts, that with a few exceptions, they may be said to be well supplied with the means of education, and that there is scarcely an individual who has not been taught to read.

The remaining six synods, however, namely, Argyle,

* *(Sinclair's Survey.)* Martin mentions, at the beginning of last century, that Macneill of Barra, and his followers, were catholics, and that he exercised considerable influence.

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Glenelg, Ross, Sutherland, and Caithness, Orkney, and Zetland, situated chiefly in the Highlands and Islands, and containing only 143 parishes, and a population of 377,730 persons, are, as stated in the parochial returns, in the most urgent need of not less than 250 additional schools.

The number of scholars that would attend each of these 250 schools, it is computed, at a low average, would amount to forty-two. It follows, therefore, that in these synods, there are 10,500 children left without any adequate means of education; and the Committee are quite satisfied that the number is, in fact, much greater than the calculated number of 10,500. These 10,500 children alluded to are all, it is to be noticed, under fifteen years of age. If persons of all ages are included, the number of those not taught to read almost exceeds belief. But how could it be otherwise, when more parishes than one are described, as not having a sufficient number of schools to accommodate one-tenth of their population? Several are said to be in need of three and four, and one of even six schools.

In the first ten synods above mentioned, there are only six catechists stated to be necessary for the due means of religious instruction to the people, and this necessity arises from the large territorial extent of some particular parishes.

In the other six synods also above mentioned, no fewer than 130 catechists are required! Nor will this lamentable deficiency seem surprising, when the physical localities of the country are considered. There are many islands in it at great distances from the coast. The coast of the main-land is often indented by long arms of the sea, and its whole surface is intersected, and in many places, rendered impassable by precipitous mountains, and by rapid rivers.

By the authority of the General Assembly, congregational collections, throughout the parishes of Scotland, were recommended, and materially augmented the fund for the formation of schools.

In November 1825, the first school was opened in the parish of Lochbroom: and in May 1827, the number of schools had amounted to thirty-five; whilst others were in progress, and the number of children and adults receiving education amounted to nearly 3000. In these schools both Gaelic and English are taught; and according to the returns received from 23 schools, 681 were reading Gaelic, and 1433 English.

In justice to the Catholics, it must be stated that they are eager for knowledge, and that in many instances, their efforts to satisfy their desire, have received the co-operation of their clergy. At Kinloch Moidart, near Arisaig, the Gaelic School Society has a seminary, respecting which it is stated in the report of 1827, "that though the school has been established here but a year, many evince an extensive acquaintance with the blessed truths contained in the Bible, which is peculiarly pleasing, as the prevailing religion in this quarter is the Roman Catholic. The parents seem deeply sensible of their obligations to your society; and some employ their children in reading to them out of the Scriptures, which heretofore were to them as a sealed book."

In the neighbourhood of Arisaig, the British and Foreign Bible Society distributed, during this year, upwards of one hundred Bibles, gratuitously, among the Catholics, on their own application. In the neighbouring islands the Gaelic School Society received no opposition; and the following extract from the *Report of the Committee of the General Assembly, for increasing the Means of Education and Religious Instruction in Scotland, by means of Schools*, is equally satisfactory. "The Sub-Committee feel cordial gratification in reporting, that one extensive Roman Catholic proprietor, has joined cheerfully in providing a portion of the required accommodations for a schoolmaster; nor can they, without injustice to the present Roman Catholic Bishop in the isles, avoid recording, that he has promised to be the convener of the Committee, with the most enlightened and liberal frankness; and to employ his influence, for encouraging the attendance of the children of the Catholic persuasion in the General Assembly's Schools." The venerable and benevolent Principal of the University of Edinburgh, who proposed, and has promoted, by his visits to the islands, this scheme for the education of the people of these regions, personally received from the above-mentioned bishop, and several priests in the Long Island and elsewhere, the frank and cordial offer of their co-operation.

SMALL ISLES. EGG.

On leaving Arisaig, we were driven by a contrary wind into Egg. Its harbour is formed by a sound, sheltered by a rocky island, and capable of admitting a vessel of seventy tons. A few cottages, and two decent houses, appear on the sloping side of an adjoining ridge. One of these is the Manse: In one day they were deprived of their possessors; the late minister and physician, who were drowned within pistol-shot of the shore. Such are the accidents to which the function of minister is liable in these islands. The last incumbents of Stornaway, in Lewis, and of a parish in Arran, perished in a similar manner. In our walk to the Manse, we experienced that sort of molestation to which pedestrians are perpetually subject in Scotland. A man of very respectable appearance accosted us, and asked us each successively respecting the names of the others of our party, requesting us to inform him accurately, as he should be questioned concerning us by the whole island. "What is your name? your object in travelling? whence came you? whither are you going? where are you staying when you will be at home?" are customary queries. The answer by no means satisfies: suspicion questions its correctness: and the sight of a stranger, and the speculations which it suggests, often interrupt the work of a field, during the long period in which he is visible in the open country over which he is travelling. The advice of Burns is in conformity to this practice.—

> Conceal yoursel as weel as ye can from critical dissection,
> But keek thro' every other man with closest sly inspection.

The pathway to the hospitable manse was rough, crossing a broad and rapid torrent. The principal object of interest in the island, is the Scuir, the summit of which rises to an elevation of 1340 feet above the level of the sea, shooting up into columnar precipices, varying, according to Macculloch, from 470 to 350 feet in height, and composed of black porphyritic pitch-stone: a stately pile of natural architecture*. The island, seen from its summit, exhibits a wild, chaotic waste, interspersed by a multitude of small lakes, enclosed in the rugged hollows of the hills.

From the Scuir, we descended to the beach on the eastern side of the island, and visited the cave. Its entrance, choked by long grass, can be penetrated only on hands and knees. Its interior extends about 20 or 30 feet in height and breadth, and about 250 in length. It is yet strewed with bones, remnants of the black mouldering relics of the entire population of the island, who perished in it, victims of vengeance in a lawless age. The tale, horrible as it is, is well authenticated. Some clansmen of Macleod of Macleod, in Sky, were driven into Egg by contrary winds when on their return from a southern cruise, and seriously offended the natives, who had hospitably entertained them. These, resenting their conduct, turned them adrift upon the sea, in a boat without oars. The tide fortunately set towards Sky, and landed them in that island. The people of their clan, incensed at this outrage, instantly repaired to Egg to avenge the injury. The poor natives, perceiving the approach of the hostile flotilla, took refuge in the cave, and were sought in vain by the invaders, who could not discover their retreat, till unhappily, one of them who was sent out to ascertain whether the enemy had departed, was observed, and traced to his hiding-place. A fire was kindled at the entrance, and the inmates were suffocated. An aged gentleman in Sky, whose memory is richly stored with local traditions, informed me, that he had read several of the songs celebrating the achievements of the hero by whom this deed of summary vengeance was perpetrated. He was renowned for his prowess, had defeated Clanronald on the mainland, and perished in a sea-fight in Bloody Bay, in the Sound of Mull. But of all his feats, the burning the cave in Egg obtains the greatest meed of applause: an unequivocal proof of the spirit of the age. Yet some extenuation of this act of barbarous indiscriminate massacre may be supplied, by the recollection, that in the period of anarchy in which it occurred, the islanders were frequently compelled, in self-defence, to take the law into their own hands, and to inflict on each other punishments, on a principle, which, in the intercourse of states, becomes a mainspring of international security.

The harbour of Egg was much crowded when we re-

* It was first brought to notice at the commencement of the present century, by Professor Jamieson of Edinburgh. (In the first part of these sketches, the discovery is inadvertently ascribed to Playfair.)

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turned to our cutter for the night. The shore and the boats were filled with people, young and old, fishing for the *coal-fish*, which bears different names, cuddy, sithe, sillock, &c., according to the period of its growth, not attaining its full size till the fourth year of its age. It affords a plentiful harvest to the fisherman. The readiness of these and other fish to bite on this prosperous evening, was quite astonishing. The real benefit which the islanders derive from this easy substitute for more adventurous and far more lucrative sea-fishing, may be questioned. There are few people, to whom the temptation to the indolent enjoyment of slender profits, purchased at the cost of little labour, is more agreeable, and consequently more dangerous, than to these islanders: a people when fairly put upon their mettle, capable of arduous and daring exertion, but ready to grasp at every pretext for avoiding it, when not absolutely necessary. The coal-fish furnishes not only food, but oil to the rude lamp of the Highlander. Thus, according to the benignant economy of Providence, the ocean supplies, at his own door, the materials for illuminating the page, from which the light of fancy, of reason, or of religion, beams on the mind of the imaginative and intelligent native of these dark and stormy regions.

Egg, Rum, Muck, and Canna, form the group called the Small Isles. Egg contains 6000 acres, of which one sixth is arable. Its population has been diminished by emigration. These islands constitute one parish, served by two ministers, one of whom is a missionary. The Catholic priest who officiates in Egg and Canna, resides in the former. It must be regretted, that the School of the Gaelic School Society has been discontinued, that institution shifting the position of its schools, in order to diffuse instruction to the utmost extent of its funds.

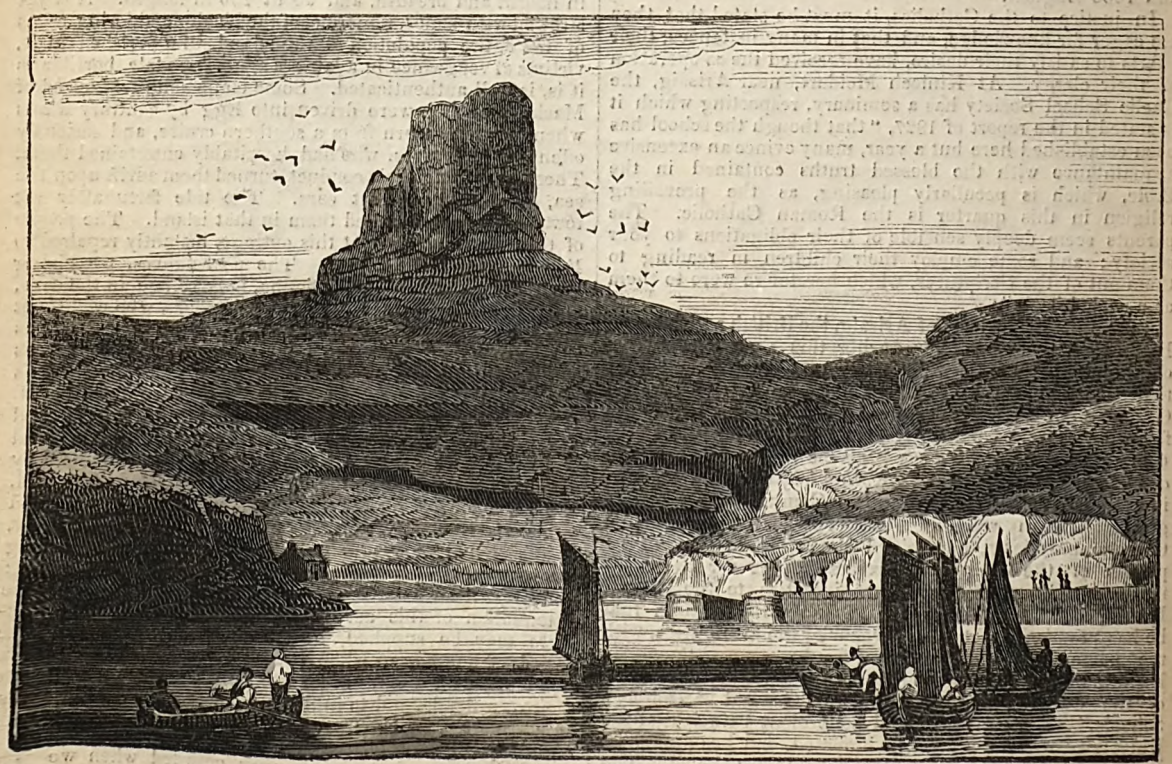
Rum, a huge pile of mountains, is famous for its breed of horses, originally planted here by a vessel belonging to the Spanish Armada, and for the production of a beautiful blood-stone. To the mountains of Rum Sir Walter Scott has most appropriately applied the epithet "dark." Of this island Dr. Macculloch observes: "There is a great deal of stormy magnificence about the lofty cliffs, as there is generally all round the shores of Rum; and they are in most places as abrupt, as they are inaccessible from sea. The interior is one heap of rude mountains, scarcely possessing an acre of level land. It is the wildest and most repulsive of all the islands. The outlines of Halival and Haskeval are indeed elegant, and render the island a beautiful and striking object from the sea. In some places, extensive surfaces of bare rock are divided into polygonal compartments, so as to resemble the grand natural pavements of Staffa, but with an effect infinitely more striking. If it is not always bad weather in Rum, it cannot be good very often, since, on seven or eight occasions that I have passed it, there has been a storm, and on seven or eight more in which I have landed, it was never without expecta tion of being turned into a cod-fish."

The mountainous islands among which Rum is prominent, are, in fact, barrier crags which intercept the vapours of the Atlantic. Thus Ireland, to borrow the peculiarly poetical metaphor of a physician residing in Cornwall, who has written a book in commendation of its climate, serves as "an umbrella" to that fortunate extremity of our island. But to the difference of climate resulting from local circumstances, we shall have occasion to advert hereafter; a subject on which the pedestrian can offer the testimony of personal experience.

The principal curiosity in Canna, is a rock possessing magnetic influence, which affects the compass. "But such disturbances," says Macculloch, "are neither peculiar to that point, nor even to this island. Deviations of the needle, produced by the influence of rocks or land, are very frequent throughout all the basaltic islands of this coast; and, in many places, the influence is such, and so extensive, as to affect the ordinary variation of the compass when at sea." The scenery of this island is highly picturesque.

Muck yields good pasturage. This island frequently suffers extremely from the want of fuel, occasioned by the deficiency of peat, a source of wealth and comfort, the value of which can be only fully appreciated when severely felt. The peat is conveyed to Muck from the main-land in boats, often so heavily laden as to risk being swamped. We found Egg subject to a similar inconvenience, caused by the perpetual rain, which had prevented the drying of the peat. The more prompt and industrious among the highlanders will not, however, admit this excuse, and assert that whenever the people are quick in seizing the opportunity, they infallibly secure a good dry stock of fuel. On the opposite coast the peat is piled under the roofs of the houses. It is thought that the increased facilities for procuring coal, would enable the people to purchase it often at a cheaper rate than peat.

P. S. Q. R.



THE SCUIR OF EGG.

LONDON: Published by JOHN WILLIAM PARKER, West Strand; and sold by all Booksellers.

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THE

Saturday Magazine.

No. 140. SEPTEMBER 6TH, 1834. Price One Penny.

UNDER THE DIRECTION OF THE COMMITTEE OF GENERAL LITERATURE AND EDUCATION APPOINTED BY THE SOCIETY FOR PROMOTING CHRISTIAN KNOWLEDGE.

EVE'S APPLE, OR THE FORBIDDEN FRUIT.

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THE FORBIDDEN FRUIT,

OR, EVE'S APPLE TREE OF CEYLON.

THE island of Ceylon is situated between the sixth and tenth parallels of North latitude; winter is consequently unknown, and it enjoys a summer which may be styled perennial. The richness and variety of its natural productions are indescribably great, though it is remarkable that the soil of the country contains a very small proportion of vegetable matter; a fact attributable, probably, to the high temperature of the climate *, which produces rapid decomposition, and to the heavy rains which prevent its accumulation. The soils, according to Dr. Davy, are derived from the decomposition of gneiss, of granitic rock, or of clay iron-stone; and in many cases, quartz constitutes more than nine-tenths of the whole. Of this the principal cinnamon-garden, in the neighbourhood of Colombo, is a singular instance; in many places the surface of the ground is as white as snow, being a pure quartz-sand: a few inches below the surface, where the roots of the plant penetrate, it is of a gray colour, and, upon being analyzed, was found to consist of more than ninety-eight parts of siliceous sand, to one part of vegetable matter. It would appear, therefore, not a little surprising, that cinnamon should succeed best on so poor a soil, but this success is attributable to the operation of other circumstances.

The peculiarities of the climate of Ceylon cause its productions to differ very much on the opposite coasts; the Palmyra-tree *(Borassus flabelliformis,)* for instance, which is extensively cultivated on the northern side, is hardly to be met with on the south; while, on the contrary, the Coco-tree, *(Cocos nucifera)*, which forms a sort of continuous garden in the south, cannot be grown on the north: in fact, all vegetable productions requiring a moist soil succeed best on the south-west side, and those requiring a dry one, on the north and north-east sides of the island. The seeds of all European plants degenerate very much, and, in a few years, yield but very indifferent returns: to preserve the quality, the importation requires to be renewed almost every year; but some of the indigenous plants flourish with wonderful vigour.

Dr. Davy remarks, that the geology and mineralogy of Ceylon, have not yet received that attention which their importance demands; the same observation is fairly applicable to the other branches of the natural history of this beautiful and valuable possession of the British Empire. We are thus unfortunately precluded from giving any very detailed description of the botanical curiosity, of which we this week present an engraving, executed from original drawings, kindly furnished to us with that view, by the Right Hon. Sir Alexander Johnston, who, while inquiring into the history of the island, had drawings made of a great many of the trees, plants, and other vegetable productions, to which any religious, political, or moral interest was attached by the native Hindús, Buddhists, Mohammedans, or early Christians. To that gentleman is to be ascribed, amongst other plans for the improvement of the island of Ceylon, and the developement of its resources, the suggestion, in the year 1810, of establishing a botanical garden near Colombo, which was accomplished, and of having the Linnean system translated into the Singhalese and Tamul languages, and some of the ablest of the natives regularly instructed in botany. We are indebted to Mr. Moon, the late superintendent of the garden, for having arranged, according to that system, a valuable catalogue of Ceylonese plants, in the English and Singhalese languages.

The subject of our sketch occurs in this catalogue, as the *Tabernæmontana dichotoma* of the Hortus Kewensis. Its native name is *Diwi Kaduru*, and nine species of the tree are enumerated. *Kaduru* signifies "forbidden," and *Diwi* "tiger's." It thrives in a low situation with a light mixed soil, and is found near Colombo.

The flower of this extraordinary production is said to emit a fine scent; the colour of the fruit, which hangs from the branches in a very peculiar and striking manner, is very beautiful, being orange on the outside, and a deep crimson within; the fruit itself presenting the singular appearance of having had a piece bitten out of it. This circumstance, together with the fact of its being a deadly poison, led the Mohammedans, on their first discovery of Ceylon (which they assigned as the site of Paradise), to represent it as the "forbidden fruit" of the garden of Eden; for, although the finest and most tempting in appearance of any, it had been impressed, such was their idea, with the mark of Eve's having bitten it, to warn men from meddling with a substance possessing such noxious properties †.

The traditions which connect the history of our first parents with various localities, both in Ceylon and other eastern islands, are of such ancient date, that their origin becomes a subject of curious speculation. Adam is represented by the Moormen, or Mohammedans of Ceylon, on his expulsion from Paradise, to have lamented his offence, standing on one foot on the summit of the mountain which now bears his name; the figure of a foot is still to be traced there, but this, the Buddhists claim as a relic of their deity. Again, the reef of rocks connecting Ceylon with the island of Rámiseram, is usually called Adam's Bridge; but the Hindús, on the other hand, term it Rámá's bridge, representing their hero to have crossed it, when about to attack the giant Rávana in his strong-hold; and two large monuments, like Mohammedan tombs, on the island of Rámiseram, are represented by the Mohammedans to be the tombs of Cain and Abel. Many other instances might be adduced, but they may be more properly traced at another opportunity, with the aid of the additional information which Oriental scholars are so actively engaged in gathering for the illustration of ancient history.

* The following results are given by Dr. Davy as the annual mean temperature at the different places indicated, viz.,

Trincomalé, on the N.E. coast	80° 4′
Colombo, on the S.W. coast	79 0
Kandy, in the interior	73 5

The summit of Adam's Peak, about 7000 feet above the sea, varied from 51° at 9 P.M., to 59° at 7 A.M., in April, 1817.

† We find that these particulars were furnished by Sir. A. Johnston to Captain Grindlay, in illustration of a view of the cinnamon-garden from the back of Sir Alexander's house, published by Captain G. in his beautiful work on the scenery, architecture, &c., of Western India and Ceylon.

No one in his heart derides religion long. What are we—any of us? Religion will soon be our only care and friend. —PALEY.

No sound should be heard in the church but the healing voice of Christian charity. The cause of civil liberty and civil government gains as little as that of religion by confusion of duties. Those who quit their proper character, to assume what does not belong to them, are, for the greater part, ignorant both of the character they leave, and of the character they assume. Wholly unacquainted with the world in which they are so fond of meddling, and inexperienced in all its affairs, on which they pronounce with so much confidence, they have nothing of politics but the passions they excite. Surely the church is a place where one day's truce ought to be allowed to the dissensions and animosities of mankind.—BURKE.

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ON HABITS.

Considering how deeply the principle of imitation is implanted in human nature, and how intimately it is connected with all that we say, or think, or do; and that the effect of this principle is to establish an undistinguishing rule of action, and an indolent subservience to custom, we need not be much surprised to find man, too frequently, the mere slave of habit.

From his earliest infancy, indeed, he is occupied, almost unconsciously, in forming a system of habits, by which his daily life is regulated, and on which, more or less, his future happiness or misery depends. It can scarcely, however, be supposed, that judgment or foresight have much, if any, part, in these first beginnings; and the result must mainly depend on the associations presented to the mind of each individual, and the character of those by whom he is surrounded. In maturer years, when this system comes to be submitted to the test of a strict examination, and to be reformed by the rule of experience, often dearly bought, a task is frequently to be commenced, which it requires all the courage of the most stout-hearted to engage in; and happy is he who, when he shall have detected the defects of the structure which has been thus almost imperceptibly raised, can set himself resolutely, and without hesitation, to pull down all that is unsightly, useless, or pernicious in its parts, even although the foundation itself is involved. Having formed his resolution, let him that instant set about the execution of it, and let him rest assured he will have occasion for all the energies he can bring to the task. If it be true that a freedom from error, and an honest reception of truth, constitute our happiness in this life, as well as our title to a better, it will follow that no sacrifice we can make in the pursuit of such an object can be too great; and if we proceed upon a well-grounded conviction that there is no *uncertainty* in the matter, it will also follow that the exertion must, in course of time, succeed.

But alas! We are too willing to put off the evil day, and while we are dallying with every trifle in our path, instead of resolutely girding ourselves up to the effort, time steals on, and life languishes, at best only a succession of failures which serve but more strongly to rivet our chain.

To say nothing of the acknowledged consequences of open and gross vice, how many a well-intentioned scheme do we not see frustrated by some circumstance which we readily impute to chance or accident, but which, if the truth were told, may be fairly traced to a weakness, an indecision, a want of something which we have long surrendered to the tyranny of habit, and which we have not the courage to reclaim. An undistinguishing submission to the customs of the world, a dread of its censure, and an acute sensibility of its applause; an indolent acceptance of the plausibilities of error, and an aversion to the investigation of abstract truth; are fetters which we are too willing to permit to be thrown around us, and under the constraint of which, many a good purpose languishes and dies. Yet, so satisfied are we with our bonds, so corrupt is our nature, so perversely indolent, yet so sensitively proud, there is no self-deception we would not practise, rather than encounter the evil, no error we would not fall into, rather than admit its existence.

I never think of this subject without recurring to the instructive story of *Obidah and the Hermit* in *The Rambler*. Obidah is described as setting forward on his journey through the plains of Indostan; his senses are regaled by all that could delight or amuse him, and forgetting the toils of his journey, he steadily pursues his path till the sun reaches its meridian height. At a little distance from the high road he espies a shady-grove, which offers a tempting invitation to screen him from the glare of the mid-day sun; he enters, and, enchanted with the spot, strolls incautiously on, amusing himself with plucking flowers and listening to the sound of birds. In these amusements his hours pass away unnoticed; having wandered from the direct path, he knew not which way to travel, and he stood pensive and confused, afraid to go forward lest he should go wrong, and yet conscious that his time for loitering had long since expired. The day draws to a close. A storm comes on, and, seeking for shelter, he arrives unexpectedly at the cell of a Hermit; his story is told, and the hermit takes occasion to read him an instructive lesson, which it will be well for us all to ponder and apply to ourselves.

"Human life," says he, "is the journey of a day; temptation succeeds to temptation, and one compliance prepares the way for another; we lose in time the happiness of innocence, and we solace our disquiet with sensual gratifications. We entangle ourselves in business, immerge ourselves in luxury, and rove through the labyrinth of inconstancy, till the darkness of old age begins to invade us, and disease and anxiety obstruct our way. We then look back on our lives with horror, grief, and repentance, and wish, too often vainly wish, we had not forsaken the ways of virtue and happiness.

"Happy are they who shall learn from thy example not to despair, but shall remember that though the day is past, and their strength wasted, there yet remains one effort to be made; that reformation is never hopeless, nor sincere endeavours ever unassisted; that the wanderer may at length return after all his errors, and that he who implores strength and courage from above, shall find dangers and difficulties give way before him." H.

THE COMMON BEE.

The Common Bee, or Honey-Fly, is an insect of the species of the fly with four wings. This fly is of the number of those who live in association. Man has subjected them to his dominion, in order to profit by their labour; and he has assembled them in kinds of baskets, or boxes, called hives, which vary in form and size in different countries.

The Bees live in a state of society; the individuals of a hive are perfectly known to each other, and they never admit a stranger into their community, excepting accidentally at swarming-time, when circumstances can so combine, that several swarms may unite and form a social brotherhood. Every society is a monarchy governed by a queen, subordinate to whom are several hundred drones, and a multitude of labourers, according to the size of the colony. It is of the latter that we are giving an account.

These insects are called common, because they, in fact, compose the community of the hive, of which the drones only form a part during a short period of time. They are also called Working Bees, because they alone bring provisions into the hive, construct the combs, nourish the brood, defend the monarchy; in one word, because they perform all the labour useful to the community.

Some authors maintain that, in the monarchy of the Bees, a regularity and an admirable subordination are to be observed; that a well-regulated distribution of employment is remarked, as well as perfect order and concert, which must result from minds

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conspiring to the execution of the same plan; but that which in men would be the effect of reason, correspondence, or co-operation, is in the Bees but the effect of that instinct which is implanted in them by the great Creator.

We are acquainted in England but with one sort of Bees, although the foreign naturalists mention three, and some even four; but this latter kind is very rare, and has not yet been naturalized.

It is to this small, but wonderful insect, that we are indebted for all the honey and wax which form a part of our domestic and commercial relations. When we consider that the former is amassed from those small, and to us almost imperceptible, globules which are found either in the chalice of the flowers, or exude from the trees, we cannot be sufficiently impressed with admiration at the perseverance and labour of the Bee. It appears to labour less for the preservation of its own existence, than for that of its species, and the prosperity of its populous state! The days on which the honey abounds in the flowers, and on the leaves of certain trees, the Bee is observed to be uncommonly industrious, entering and leaving the hive with wonderful rapidity. The office of collecting the farina from the plants is not, however, neglected; and it is very easy to discriminate between the Bee which has been collecting honey, and that which has been collecting only farina. The shape of the former is cylindrical, that of the latter oval.

In regard to the physical description of the Bee, the most remarkable parts of it are the head, the breast, and the belly. On the former are observed two *rete mirabile* eyes placed in the side, two *antennæ*, two hard teeth or jaws, which play, on opening or shutting, from the left to the right. These teeth enable it to collect the wax, to knead it, to construct the cells, and to remove from the hive every obnoxious thing.

Below these two teeth we observe a proboscis, which has the appearance of a thick fleshy substance, of a very shining and chesnut colour. This substance is divided into two parts, very supple at the end, and it is only seen at its full length when the Bee is employed in collecting honey, or sipping water. This proboscis is a most wonderful machine. To the simple view, it appears enveloped with four kinds of scales, which form together a channel by which the honey is conveyed. The proboscis, which is in this channel, is a muscular body, which, by means of its muscular motions, makes the honey ascend into the gullet. If the teeth be separated, we observe, at the orifice of the proboscis, an opening, which is the mouth, and above it a fleshy substance, which is the tongue. The breast is attached to the head by a very short neck; it carries four wings on it, the two last of which are longer than the other. It has six feet, on the two hinder of which are two triangular cavities, in which the Bee, by degrees, collects the particles of farina from the plants. At the extremity of the six feet are two sorts of fangs, with which the Bees attach themselves to the sides of the hive, and to each other. From the middle of these fangs, on the four hinder legs, project four bushy substances, the use of which is to collect the dust of the flowers attached to the hair of their body. These brushes have the same use as hands.

The body, properly so called, or the belly, is united to the breast by a species of thread, and is composed of six scaly rings. The whole body of the Bee appears, even to the naked eye, to be well clothed. Age makes a little difference in them, in point of colour; those of the present year are brown, and have greyish hair; those of the preceding year have reddish hair, and the scales less brown, rather inclining to black; their wings are also often torn and fringed at the ends, by their former flights. On the breast, and on the wings of the body, are observed small orifices or pores, in the shape of a mouth, by which the Bee respires; these are the lungs, and they are technically called *stygmates;* this part, which is of a wonderful construction, is common to them, as to all other insects.

The interior of the belly consists of four parts,—the intestines, the honey-bag, the venom-vessel, and the sting. The intestines serve for the digestion of their food. The honey-bag, when it is filled, is as large as a small pea, transparent as crystal, and contains the honey which the Bees have collected from the flowers, and which is disgorged into the cells to nourish the hive during the winter. That which is destined for their own nourishment never enters into it, but passes through the viscera destined to the purpose of digestion. The vessel which contains the venom is at the root of the sting, along which the Bee ejects some globules, as along a tube, in order to spread into the wound. The sting is situated at the extremity of the belly of the Bee; it is about two lines in length, and enters with great rapidity, by means of certain muscles which are placed very near the sting, and which are very perceptible on squeezing the hinder part of the Bee; its extremity is barbed, the teeth of which are turned in the direction of an arrow, which enter with facility, and cannot be extracted without causing a laceration. The wound which the Bee makes is almost always fatal to it; when it wishes to withdraw its sting, it remains in the wound, and with it the Bee loses the vessel of venom, which is at the root of the sting, and the ligaments to which it is attached. The Bee thus wounded cannot live a long time; it perishes, after having made war, in the manner of the savages, with poisoned arrows.

These details can only produce, in every rational man, a more distinct and extensive knowledge of that infinite intelligence, which has arranged the creatures of this earth, presided at their organization, and regulated their existence and configuration. There is nothing in nature which can so forcibly demonstrate to us an equally wise and powerful Author. The insects the most vile are, perhaps, more admirable in their construction than the sun and the most brilliant stars. What proportion! what harmony! what correspondence, in every part of the Bee! How many combinations, arrangements, causes, effects, and principles, which tend to the same end, and concur in the same design! What exactness, what symmetry in its little body, apparently contemptible, and so little admired by ignorant and inattentive persons! As in the greater number of animals, so we observe in the Bee, vessels without number, liquids, motions often united in an imperceptible point,—all the organs of life, the instruments of labour, the means of escaping from their enemies, weapons to command victory, and a thousand beauties which adorn its exterior form!

Every thing in these insects announces that supreme wisdom which presided at the formation of a work, so perfect, so industrious, so superior to every thing which art could ever produce. Every thing here is for our use and benefit. The Bees, in fact, make use of their wondrous qualities only for our good. It is for us that they work; and it is towards Him, therefore, who has given to them these inclinations, that we ought to express our love and gratitude.—HUISH *on Bees.*

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HENRY PRINCE OF WALES; ELDEST SON OF KING JAMES THE FIRST.

PRINCE HENRY, SON OF JAMES THE FIRST.

THERE are few events recorded in the history of England, which the generality of readers, the young especially, peruse with so much interest as the early death of illustrious and promising characters. The pen of the impartial historian dwells with delight on those traits of disposition, which gave rise to the fairest hopes; and imparts to the reader emotions of sorrow and regret, at the premature close of a life, dear to thousands. It is not in a political point of view, however, that a great national loss is *thus* felt. Honour, love, and esteem for the individual character, must be the spring of such affections,—pensive indeed, yet mingled with pleasure, that so sweet a plant was removed to a kindlier soil, before the rude breath of the world had disturbed or corrupted it; and that one of lofty station left a pattern, which in its leading points, all, however lowly in their walk of life, may follow, and be happy.

Deep and universal was the grief which pervaded the nation, on the decease of the young prince, whose likeness appears at the head of this paper. Born to high expectations, and surrounded by busy flatterers, HENRY, PRINCE OF WALES, had established a name for piety, temperance, prudence, and many manly virtues, when he was snatched away in the very spring-time of existence! Yet he had the happiness to die in the height of favour with men, as well as, we hope, with God, and without experiencing the miseries which awaited the royal family. Of his high qualities and exemplary behaviour, several authentic documents exist; and the scattered intelligence concerning him, appears well worthy of collection for our Magazine,

HENRY, eldest son of King James the First, and Queen Anne of Denmark, was born in Scotland, early in 1594. After remaining under the care of the Earl of Mar, he was placed, at five years of age, with an excellent tutor, by whose instructions he made a great and rapid progress in learning; the energies of the body keeping pace with those of the mind. At the age of nine he began to acquire a fondness for riding, dancing, shooting, and tossing the lance, exercises in which he afterwards greatly excelled; and before he reached the age of ten, he was installed at Windsor, a knight of the garter. On the 4th of June, 1610, he was created Prince of Wales, at Westminster, with solemn and magnificent ceremonies, the eyes of the people being fixed upon him as their future sovereign.

It was now that he became most popular among all classes. In the government of his affairs, he set a noble example of mingled liberality and economy; providing plentifully, but knowing and watching his expenses; and though with a retinue of little less than five hundred persons, many of them young gentlemen of high expectations, he left his revenue increased some thousands a year. An original manuscript, containing orders made by this young prince, respecting his household, as given at Richmond in 1610, was communicated some years ago to the Royal Society. The first order is; "That when I am at divine service in my private closet, my gentlemen in ordinary be warned to attend me, and be present at times of prayer; and to do the like when I go to my public chapel to service and sermons: wherein I will dispense with no man; holding him

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unfit to serve me, that with me will forbear to go to hear the word of God; which example of liberty shall never be tolerated in my court, nor made a reason to encourage others in like disobedience and contempt towards religion."

After a series of general regulations concerning his own and his household's living, he concludes, "As I began with the due divine service unto Almighty God, without which nothing can prosper nor yield comfort, either in this world or in the world to come; so do I conclude, that amongst other my ordinances, it be strictly looked unto and observed that, four times in the year, namely, at Christmas, Easter, Midsummer, and Michaelmas, all my ordinary servants, without exception, do receive the Communion at my public chapel; and that before the receiving of the communion, one of my chaplains, or some other good preacher, do make a sermon, or read a lecture, tending to instruct men to the reverent and worthy receiving of that holy and blessed Sacrament. And of such as shall either wilfully refuse so to do, or cautiously absent themselves of purpose, I desire that myself be informed, to give such further order therein, as may stand for an eminent example and chastisement to such ungodly and unchristian-like disposition: for the which kind of people, my court shall be no shelter, nor my service any protection."

He had an esteem for the brave and unfortunate Sir Walter Raleigh, during whose sad imprisonment, the prince used to say, *Sure no king but my father would keep such a bird in a cage!* In an interesting letter of advice to the prince, from Sir Walter, dated August, 1611, among other excellent passages we find these: "Consider the inexpressible advantage which will ever attend your Highness, while you make the power of rendering men happy, the measure of your actions. While this is your impulse, how easily will that power be extended! The glance of your eye will give gladness, and your very sentence have a force of bounty." And his royal father, who, it is asserted, sometimes felt himself outdone by the splendour of the prince's reputation, addressed to him the following powerful lines.

FROM KING JAMES TO PRINCE HENRY.

GOD gives not kings the style of Gods in vain,
For on HIS throne his sceptre do they sway:
And as their subjects ought them to obey
So kings should fear and serve their GOD again.

If then ye would enjoy a happy reign,
Observe the statutes of our heavenly King:
And from his law make all your laws to spring;
Since his lieutenant here should ye remain.

Reward the just; be stedfast, true and plain;
Repress the proud, maintaining aye the right;
Walk always so, as ever in HIS sight,
Who guards the godly, plaguing the profane.

In 1612, a marriage was proposed for him with a daughter of Henry the Fourth of France. But, though not at once rejecting, he never appeared desirous to encourage this union, on account of the princess's creed, she being a Roman Catholic: and it is stated that "in his sickness afterwards, he applied this chastisement for a deserved punishment upon him, for having ever opened his ears to admit treaty of a popish match." In October, the same year, he was seized with an illness, the nature of which was not thoroughly understood; and he died, deeply lamented, on the 6th of November, 1612*.

* It is remarkable, that another rising Hope of England, the Princess Charlotte, was snatched away in the same month, and on the same day of the month. She died November 6, 1817.

An opinion prevailed at the time, that he was carried off by poison; a presumption not to be wondered at, nor indeed, perhaps, groundless, when we consider his honest and avowed dislike to the wretched court-minion Car, Lord Rochester, afterwards Earl of Somerset, as well as to the Howards, with an infamous branch of which family Car had united himself by marriage.

"HENRY," says Birch, in his *Lives of Illustrious Persons*, "was about five feet eight inches tall; of a strong and handsome frame, an amiable countenance, his hair auburn, and his eyes fine and piercing. He was sober, chaste, temperate, religious. He was never heard to swear, though the example of his father, and of the whole court, was but too apt to corrupt him in that respect. He took great delight in the conversation of men of honour; and those who were not reckoned such, were treated with no attention at his court. He was naturally gentle and affable; though he had a noble stateliness without affectation, which commanded esteem and respect. He showed a warlike taste in his passionate fondness for martial exercises. A French ambassador coming to take leave of him, found him tossing a pike; and asked him whether he had any commands to France: 'Tell your master,' said the prince, 'how you left me engaged.' He was eighteen years old when he died: and no historian has cast the least stain upon his character."

Sir Charles Cornwallis, treasurer of Henry's household, thus concludes an account of him; "God seeing it good to bestow another Crown upon him, excelling all that on earth was to be had or hoped: after some five days' sickness, endured with patience, and as often recognition of his faith, his hopes, and his appeals to God's mercy, as his infirmity, which affected him altogether in his head, would possibly permit; he yielded up the ghost at St. James's, next Westminster, and was interred at Westminster, where his body now resteth." M.

As rivers, when they overflow, drown those grounds, and ruin those husbandmen, which, whilst they flowed calmly betwixt their banks, they fertilized and enriched; so our passions, when they grow exorbitant and unruly, destroy those virtues, to which they may be very serviceable whilst they keep within their bounds.—BOYLE.

IT is useful to observe, in our progress through life, the chain of duties, trials, and blessings, which imperceptibly conduct us from one period to another; and how successive comforts and blessings spring from previous duties. Thus the diligence, sobriety, and virtuous habits of youth, will, in middle age, ensure to us, through God's blessing, the respect of the world, and success in our pursuits, and the active and useful employments of that period, added to early and continued piety and benevolence, will produce an old age of comfort and consolation. Thus proceeding in the way we should go, we reap, from the same source, our reward for the past, and our encouragement for the future. —MRS. KING.

IN this world we are children standing on the bank of a mighty river. Casting our eyes upward and downward, along the channel, we discern various windings of its current; and perceive that it is now visible, now obscure, and now entirely hidden from our view. But being far removed from the fountain whence it springs, and from the ocean into which it is emptied, we are unable to form any conceptions of the beauty, usefulness, or grandeur of its progress. Lost in perplexity and ignorance, we gaze, wonder, and despond. In this situation, a messenger from heaven comes to our relief, with authentic information of its nature, its course, and its end; conducts us backward to the fountain, and leads us forward to the ocean. This river is the earthly system of providence: the Bible is the celestial messenger: and Heaven is the ocean in which all preceding dispensations find their end.—DWIGHT.

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THE DAISY IN INDIA.

THRICE welcome, little English Flower!
My mother-country's white and red,
In rose or lily, till this hour,
Never to me such beauty spread:
Transplanted from thine island-bed,
A treasure in a grain of earth,
Strange as a spirit from the dead,
Thine embryo sprang to birth.

Thrice welcome, little English Flower!
Whose tribes beneath our natal skies
Shut close their leaves while vapours lower;
But, when the Sun's gay beams arise,
With unabash'd but modest eyes
Follow his motion to the west,
Nor cease to gaze till daylight dies,
Then fold themselves to rest.

Thrice welcome, little English Flower!
To this resplendent hemisphere.
Where Flora's giant-offspring tower
In gorgeous liveries all the year:
Thou, only Thou, are *little* here,
Like worth unfriended or unknown,
Yet to my British heart more dear
Than all the torrid zone.

Thrice welcome, little English Flower!
Of early scenes beloved by me,
While happy in my father's bower,
Thou shalt the blithe memorial be;
The fairy sports of infancy,
Youth's golden age, and manhood's prime,
Home, country, kindred, friends—with thee
Are mine in this fair clime.

Thrice welcome, little English Flower!
I'll rear thee with a trembling hand:
O! for the April sun and shower,
The sweet May-dews of that fair land,
Where Daisies, thick as starlight, stand
In every walk!—that here might shoot
Thy scions, and thy buds expand,
An hundred from one root!

Thrice welcome, little English Flower!
To me the pledge of Hope unseen;
When sorrow would my soul o'erpower
For joys that *were*, or *might have been*,
I'll call to mind, how—fresh and green,
I saw thee rising from the dust,
Then turn to heaven with brow serene,
And place in God my trust.

FAMILIAR ILLUSTRATIONS OF EXPERIMENTAL SCIENCE.

No. IV. HEAT. TEMPERATURE. RADIATION. CONDUCTION.

IN the present state of our knowledge, it is impossible to determine whether heat should be regarded as a substance, endowed with extraordinary powers, by which it penetrates and diffuses itself among the particles of every other element; or as a quality, inseparable from matter, and dependent on certain conditions for those unceasing fluctuations which constitute its most remarkable phenomena.

The resistless energies of this omnipotent and all-pervading agent are in constant operation. There is not an instant of time that heat is not performing some important duty in fulfilment of the Divine purposes. Among all the works of God, we know of none on which the evidences of design are more conspicuously inscribed.

Whatever may be the nature of heat, be it a peculiar *substance*, or a peculiar *property*, we know that it exists. To its influence we are indebted for the due performance of all the functions of life, for all that cheers the eye, delights the ear, and gratifies the taste. Nor is it to heat only, but to its being supplied to us in its due proportions, that we owe so much. Its excess, or its deficiency, would be equally fatal to vegetable and animal existence. In one case, the earth would become a parched desert, in the other, an ice-bound plain.

It is important that we should distinguish between heat itself, and the sensation of heat. The first is a cause, the second its effect. With a view to prevent mistakes, by the frequent interchange of terms, meaning sometimes one thing, and at other times another, the term *caloric* is now extensively employed by scientific writers, to denote that condition of bodies, by which the sensation of heat is produced, or, in other words, to define the cause of heat, as distinct from its effects. Wishing to refrain, as much as possible, from scientific phraseology, we shall restrict ourselves to the ordinary term (heat), requesting our readers to remember that, unless the contrary is distinctly stated, it always means heat, as an element, residing in, or operating upon, matter, without any regard to our feelings.

By the continual use of the terms *heat* and *cold*, in the affairs of common life, we sometimes employ the latter term, as if it was descriptive of an element, or agent, equally energetic in its effects as any other with which we are acquainted, but whose properties are directly the opposite of those possessed by heat. Cold is only the absence of heat. It is easier, and, because we are accustomed to it, more natural to say, "It is cold," than it is to describe that condition by saying, "There is a deficiency of heat." The latter, however, is a correct definition. We know by experience, that the gradual abstraction of heat from a body, which at first may possess so much of it as to be unapproachable, induces the sensation we denominate cold. But cold is only a relative term. We know nothing of matter where heat is not present. There is less heat in one substance than in another; but of absolute cold we have no conception.

Temperature is a term that will very often occur whilst treating of the properties of heat. We think it right at once to explain its signification. The temperature of a body means its *sensible heat*, that is, the heat of which some estimate may be formed by a thermometer*, a useful instrument, that we shall describe particularly hereafter. In comparing two different substances, or two distinct parts of the same substance, if we find the first communicates to the thermometer more heat than the second, we say the temperature of the former is *higher* than that of the latter, or, that the temperature of the latter is *lower* than that of the former. Higher and lower, as applied to temperature, are terms that evidently owe their origin to the operation of the thermometer; since the smaller the quantity of sensible heat present in any substance with which the bulb of a thermometer is placed in contact, the lower will the column of mercury, or other fluid within the tube, descend; the greater the quantity of sensible heat, the higher will it rise. The sensible, or as it is commonly termed, *free* heat, thus discoverable in any particular substance by the aid of a thermometer, must be viewed, as entirely independent of the heat which permanently resides in that substance, or which may be temporarily combined with it in a *latent*, that is, a concealed state. We may satisfy ourselves, that a vast quantity of heat has entered into some particular substance, but we can neither detect the presence, nor estimate the quantity, of that which is latent, by our ordinary perceptions, nor through the agency of a thermometer.

Heat is communicable from one substance to

* The thermometer obtains its name from two Greek words, *therme*, heat, and *metron*, a measure.

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another by *radiation* and by *conduction*. Radiation takes place between bodies whose temperatures are unequal, at sensible distances. Contact is a condition essential to conduction.

If a piece of heated metal be fixed in the centre of a room, midway between the ceiling and the floor, heat will be disengaged from it equally in all directions, upwards, downwards, horizontally, and obliquely, which may be proved by the melting of a small quantity of tallow placed at certain distances around the metal. This is an instance of radiation. When the bowl of a metal spoon is left, for a few minutes, in a cup of hot tea, the handle of the spoon acquires the same temperature as that of the tea. Here we have an instance of conduction. In one case, the heat separated from the metal will affect the tallow at some distance, passing readily through, or among, the particles of the intervening air. In the other case, the heat first communicating with that part of the spoon in contact with the tea, it is, if we may employ the expression, pushed forward from particle to particle of the metal, along the handle, until it reaches its extremity.

As radiation and conduction commonly operate together, they may be considered as different parts, or rather, different forms, of the same process; both equally dependent on that property peculiar to heat, by which it tends to diffuse itself in every direction, and among the particles of every species of matter, whatever may be its form, size, colour, or quality. Thus, if any number of vessels, some constructed of metal, others of wood, others of stone, and others of glass, each vessel containing a liquid of a different kind and at a different temperature, be placed in the same room, the liquids and the vessels containing them will, in a few hours, all arrive at the same temperature, which will be that of the air in the room. The same would, of course, be the result, with solid or aëriform bodies, as with liquids.

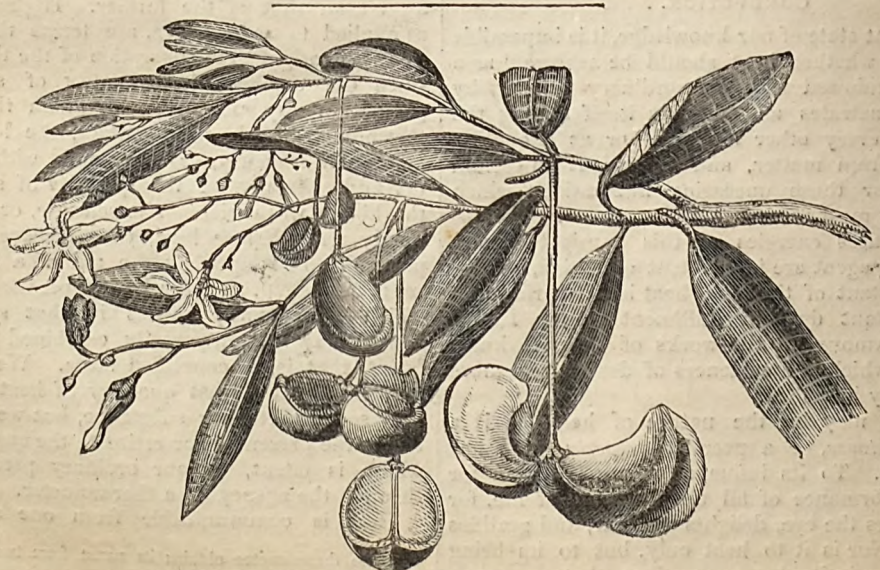
Radiation and conduction may be further explained by considering the former as operating at the surfaces of bodies, whilst the latter goes on throughout their interior parts. The rate at which heat is radiated and conducted by any substance, depends very much on the nature of the materials of which that substance is composed. Radiation is also influenced in a remarkable degree by the colours and other conditions of the surfaces of bodies.

Those bodies into which heat enters with facility, and among whose particles it is transmitted rapidly, are called *good conductors*. Those, on the contrary, which offer considerable resistance to the progress of heat among their particles, are termed *bad conductors*. The latter are frequently denominated *non-conductors*, a description not philosophically correct; since every substance with which we are acquainted will conduct heat, although in some its transmission is exceedingly slow.

Among good conductors the metals are the best: of these gold, platinum, silver, and copper, are nearly equal. The next in order are iron and zinc, then tin, and the slowest conductor of them all is lead. Wood, stone, and bricks, are among the bad conductors: of this class the most perfect are wool, hair, cotton, the fur of animals, the feathers of birds, and especially the down of the swan. Liquids and aëriform bodies, when there is no motion among their particles, are bad conductors of heat. If freedom of motion be established, they become good conductors.

In our next paper, we will endeavour to illustrate more fully the operation of Heat as respects conduction and radiation. We rather desire that our readers may complain of the brevity of our remarks, than that they should feel fatigued by our becoming tedious. R. R.

THE energy of every function is regulated in a great measure by the quantity of blood which the organs exercising that function receive. The muscles employed in the most vigorous actions, are always found to receive the largest quantity of blood. It is commonly observed that the right fore-leg of quadrupeds, as well as the right arm in man, is stronger than the left; much of this superior strength is, no doubt, the result of education, the right arm being more habitually used than the left. But still the different mode in which the arteries are distributed to the two arms, constitutes a natural source of inequality. The artery supplying the right arm with blood, first arises from the aorta, and it proceeds in a more direct course from the heart than the artery of the left arm, which has its origin in common with the artery of that side of the head. Hence it has been inferred, that the right arm is originally better supplied with nourishment than the left. It may be alleged in confirmation of this view, that in birds, where any irregularity in the action of the two wings would have disturbed the regularity of flight, the aorta, when it has arrived at the centre of the chest, divides with perfect equality into two branches, so that both wings receive precisely the same quantity of blood, and the muscles, being thus equally nourished, preserve that equality of strength, which their function rigidly demands.—DR. ROGET'S *Bridgewater Treatise*.



FRUIT AND FLOWERS OF THE EVE'S APPLE. *See* p. 90.

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